

is not the product of $1/(R_1C_1s + 1)$ and $1/(R_2C_2s + 1)$. The reason for this is that, when we derive the transfer function for an isolated circuit, we implicitly assume that the output is unloaded. In other words, the load impedance is assumed to be infinite, which means that no power is being withdrawn at the output. When the second circuit is connected to the output of the first, however, a certain amount of power is withdrawn, and thus the assumption of no loading is violated. Therefore, if the transfer function of this system is obtained under the assumption of no loading, then it is not valid. The degree of the loading effect determines the amount of modification of the transfer function.

Transfer functions of nonloading cascaded elements. The transfer function of a system consisting of two nonloading cascaded elements can be obtained by eliminating the intermediate input and output. For example, consider the system shown in Figure 3–20(a). The transfer functions of the elements are

$$G_1(s) = \frac{X_2(s)}{X_1(s)} \quad \text{and} \quad G_2(s) = \frac{X_3(s)}{X_2(s)}$$

If the input impedance of the second element is infinite, the output of the first element is not affected by connecting it to the second element. Then the transfer function of the whole system becomes

$$G(s) = \frac{X_3(s)}{X_1(s)} = \frac{X_2(s)X_3(s)}{X_1(s)X_2(s)} = G_1(s)G_2(s)$$

The transfer function of the whole system is thus the product of the transfer functions of the individual elements. This is shown in Figure 3–20(b).

As an example, consider the system shown in Figure 3–21. The insertion of an isolating amplifier between the circuits to obtain nonloading characteristics is frequently used in combining circuits. Since amplifiers have very high input impedances, an isolation amplifier inserted between the two circuits justifies the nonloading assumption.

The two simple RC circuits, isolated by an amplifier as shown in Figure 3–21, have negligible loading effects, and the transfer function of the entire circuit equals the product of the individual transfer functions. Thus, in this case,

$$\begin{aligned} \frac{E_o(s)}{E_i(s)} &= \left(\frac{1}{R_1C_1s + 1} \right) (K) \left(\frac{1}{R_2C_2s + 1} \right) \\ &= \frac{K}{(R_1C_1s + 1)(R_2C_2s + 1)} \end{aligned}$$



Figure 3–20

(a) System consisting of two nonloading cascaded elements; (b) an equivalent system.

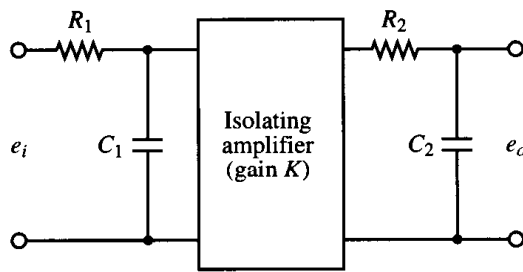


Figure 3-21
Electrical system.

3-8 LIQUID-LEVEL SYSTEMS

In analyzing systems involving fluid flow, we find it necessary to divide flow regimes into laminar flow and turbulent flow, according to the magnitude of the Reynolds number. If the Reynolds number is greater than about 3000 to 4000, then the flow is turbulent. The flow is laminar if the Reynolds number is less than about 2000. In the laminar case, fluid flow occurs in streamlines with no turbulence. Systems involving turbulent flow often have to be represented by nonlinear differential equations, while systems involving laminar flow may be represented by linear differential equations. (Industrial processes often involve flow of liquids through connecting pipes and tanks. The flow in such processes is often turbulent and not laminar.)

In this section we shall derive mathematical models of liquid-level systems. By introducing the concept of resistance and capacitance for such liquid-level systems, it is possible to describe the dynamic characteristics of such systems in simple forms.

Resistance and capacitance of liquid-level systems. Consider the flow through a short pipe connecting two tanks. The resistance R for liquid flow in such a pipe or restriction is defined as the change in the level difference (the difference of the liquid levels of the two tanks) necessary to cause a unit change in flow rate; that is,

$$R = \frac{\text{change in level difference, m}}{\text{change in flow rate, m}^3/\text{sec}}$$

Since the relationship between the flow rate and level difference differs for the laminar flow and turbulent flow, we shall consider both cases in the following.

Consider the liquid-level system shown in Figure 3-22(a). In this system the liquid spouts through the load valve in the side of the tank. If the flow through this restriction is laminar, the relationship between the steady-state flow rate and steady-state head at the level of the restriction is given by

$$Q = KH$$

where Q = steady-state liquid flow rate, m^3/sec

K = coefficient, m^2/sec

H = steady-state head, m

Notice that the law governing laminar flow is analogous to Coulomb's law, which states that the current is directly proportional to the potential difference.

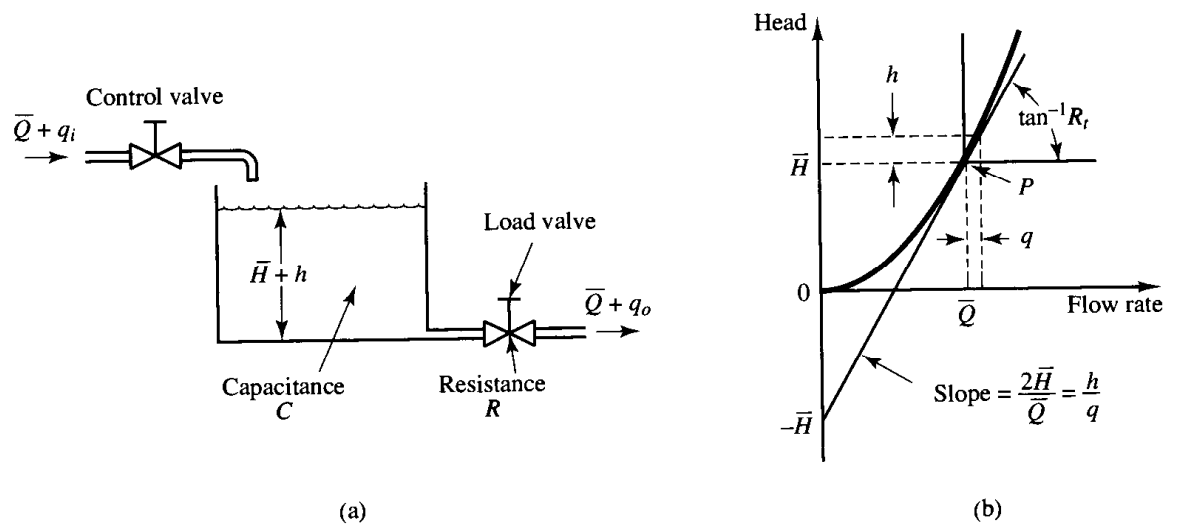


Figure 3-22
(a) Liquid-level system; (b) head versus flow rate curve.

For laminar flow, the resistance R_l is obtained as

$$R_l = \frac{dH}{dQ} = \frac{H}{Q}$$

The laminar-flow resistance is constant and is analogous to the electrical resistance.

If the flow through the restriction is turbulent, the steady-state flow rate is given by

$$Q = K\sqrt{H} \quad (3-67)$$

where Q = steady-state liquid flow rate, m^3/sec

K = coefficient, $\text{m}^{2.5}/\text{sec}$

H = steady-state head, m

The resistance R_t for turbulent flow is obtained from

$$R_t = \frac{dH}{dQ}$$

Since from Equation (3-67) we obtain

$$dQ = \frac{K}{2\sqrt{H}} dH$$

we have

$$\frac{dH}{dQ} = \frac{2\sqrt{H}}{K} = \frac{2\sqrt{H}\sqrt{H}}{Q} = \frac{2H}{Q}$$

Thus,

$$R_t = \frac{2H}{Q} \quad (3-68)$$

The value of the turbulent-flow resistance R_t depends on the flow rate and the head. The value of R_t , however, may be considered constant if the changes in head and flow rate are small.

By use of the turbulent-flow resistance, the relationship between Q and H can be given by

$$Q = \frac{2H}{R_t}$$

Such linearization is valid, provided that changes in the head and flow rate from their respective steady-state values are small.

In many practical cases, the value of the coefficient K in Equation (3-67), which depends on the flow coefficient and the area of restriction, is not known. Then the resistance may be determined by plotting the head versus flow rate curve based on experimental data and measuring the slope of the curve at the operating condition. An example of such a plot is shown in Figure 3-22(b). In the figure, point P is the steady-state operating point. The tangent line to the curve at point P intersects the ordinate at point $(-\bar{H}, 0)$. Thus, the slope of this tangent line is $2\bar{H}/\bar{Q}$. Since the resistance R_t at the operating point P is given by $2\bar{H}/\bar{Q}$, the resistance R_t is the slope of the curve at the operating point.

Consider the operating condition in the neighborhood of point P . Define a small deviation of the head from the steady-state value as h and the corresponding small change of the flow rate as q . Then the slope of the curve at point P can be given by

$$\text{Slope of curve at point } P = \frac{h}{q} = \frac{2\bar{H}}{\bar{Q}} = R_t$$

The linear approximation is based on the fact that the actual curve does not differ much from its tangent line if the operating condition does not vary too much.

The capacitance C of a tank is defined to be the change in quantity of stored liquid necessary to cause a unit change in the potential (head). (The potential is the quantity that indicates the energy level of the system.)

$$C = \frac{\text{change in liquid stored, m}^3}{\text{change in head, m}}$$

It should be noted that the capacity (m^3) and the capacitance (m^2) are different. The capacitance of the tank is equal to its cross-sectional area. If this is constant, the capacitance is constant for any head.

Liquid-level systems. Consider the system shown in Figure 3-22(a). The variables are defined as follows:

$\bar{Q}$ = steady-state flow rate (before any change has occurred), m^3/sec

q_i = small deviation of inflow rate from its steady-state value, m^3/sec

q_o = small deviation of outflow rate from its steady-state value, m^3/sec

$\bar{H}$ = steady-state head (before any change has occurred), m

h = small deviation of head from its steady-state value, m

As stated previously, a system can be considered linear if the flow is laminar. Even if the flow is turbulent, the system can be linearized if changes in the variables are kept small. Based on the assumption that the system is either linear or linearized, the differential

equation of this system can be obtained as follows: Since the inflow minus outflow during the small time interval dt is equal to the additional amount stored in the tank, we see that

$$C dh = (q_i - q_o) dt$$

From the definition of resistance, the relationship between q_o and h is given by

$$q_o = \frac{h}{R}$$

The differential equation for this system for a constant value of R becomes

$$RC \frac{dh}{dt} + h = Rq_i \quad (3-69)$$

Note that RC is the time constant of the system. Taking the Laplace transforms of both sides of Equation (3-69), assuming the zero initial condition, we obtain

$$(RCs + 1)H(s) = RQ_i(s)$$

where

$$H(s) = \mathcal{L}[h] \quad \text{and} \quad Q_i(s) = \mathcal{L}[q_i]$$

If q_i is considered the input and h the output, the transfer function of the system is

$$\frac{H(s)}{Q_i(s)} = \frac{R}{RCs + 1}$$

If, however, q_o is taken as the output, the input being the same, then the transfer function is

$$\frac{Q_o(s)}{Q_i(s)} = \frac{1}{RCs + 1}$$

where we have used the relationship

$$Q_o(s) = \frac{1}{R} H(s)$$

Liquid-level systems with interaction. Consider the system shown in Figure 3-23. In this system, the two tanks interact. Thus the transfer function of the system is not the product of two first-order transfer functions.

In the following, we shall assume only small variations of the variables from the steady-state values. Using the symbols as defined in Figure 3-23, we can obtain the following equations for this system:

$$\frac{h_1 - h_2}{R_1} = q_1 \quad (3-70)$$

$$C_1 \frac{dh_1}{dt} = q - q_1 \quad (3-71)$$

$$\frac{h_2}{R_2} = q_2 \quad (3-72)$$

$$C_2 \frac{dh_2}{dt} = q_1 - q_2 \quad (3-73)$$

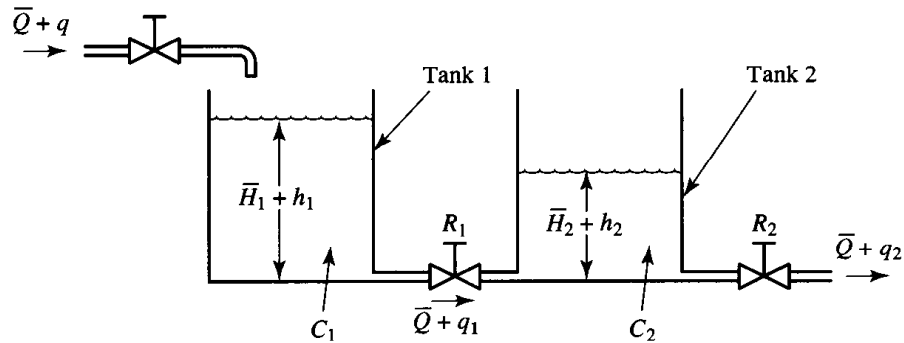


Figure 3-23
Liquid-level system
with interaction.

$\bar{Q}$: Steady-state flow rate
 $\bar{H}_1$: Steady-state liquid level of tank 1
 $\bar{H}_2$: Steady-state liquid level of tank 2

If q is considered the input and q_2 the output, the transfer function of the system is

$$\frac{Q_2(s)}{Q(s)} = \frac{1}{R_1 C_1 R_2 C_2 s^2 + (R_1 C_1 + R_2 C_2 + R_2 C_1)s + 1} \quad (3-74)$$

It is instructive to obtain Equation (3-74), the transfer function of the interacted system, by block diagram reduction. From Equations (3-70) through (3-73), we obtain the elements of the block diagram, as shown in Figure 3-24(a). By connecting signals properly, we can construct a block diagram, as shown in Figure 3-24(b). This block diagram can be simplified, as shown in Figure 3-24(c). Further simplifications result in Figures 3-24(d) and (e). Figure 3-24(e) is equivalent to Equation (3-74).

Notice the similarity and difference between the transfer function given by Equation (3-74) and that given by Equation (3-66). The term $R_2 C_1 s$ that appears in the denominator of Equation (3-74) exemplifies the interaction between the two tanks. Similarly, the term $R_1 C_2 s$ in the denominator of Equation (3-66) represents the interaction between the two RC circuits shown in Figure 3-19.

3-9 THERMAL SYSTEMS

Thermal systems are those that involve the transfer of heat from one substance to another. Thermal systems may be analyzed in terms of resistance and capacitance, although the thermal capacitance and thermal resistance may not be represented accurately as lumped parameters since they are usually distributed throughout the substance. For precise analysis, distributed-parameter models must be used. Here, however, to simplify the analysis we shall assume that a thermal system can be represented by a lumped-parameter model, that substances that are characterized by resistance to heat flow have negligible heat capacitance, and that substances that are characterized by heat capacitance have negligible resistance to heat flow.

There are three different ways heat can flow from one substance to another: conduction, convection, and radiation. Here we consider only conduction and convection. (Radiation heat transfer is appreciable only if the temperature of the emitter is very high compared to that of the receiver. Most thermal processes in process control systems do not involve radiation heat transfer.)

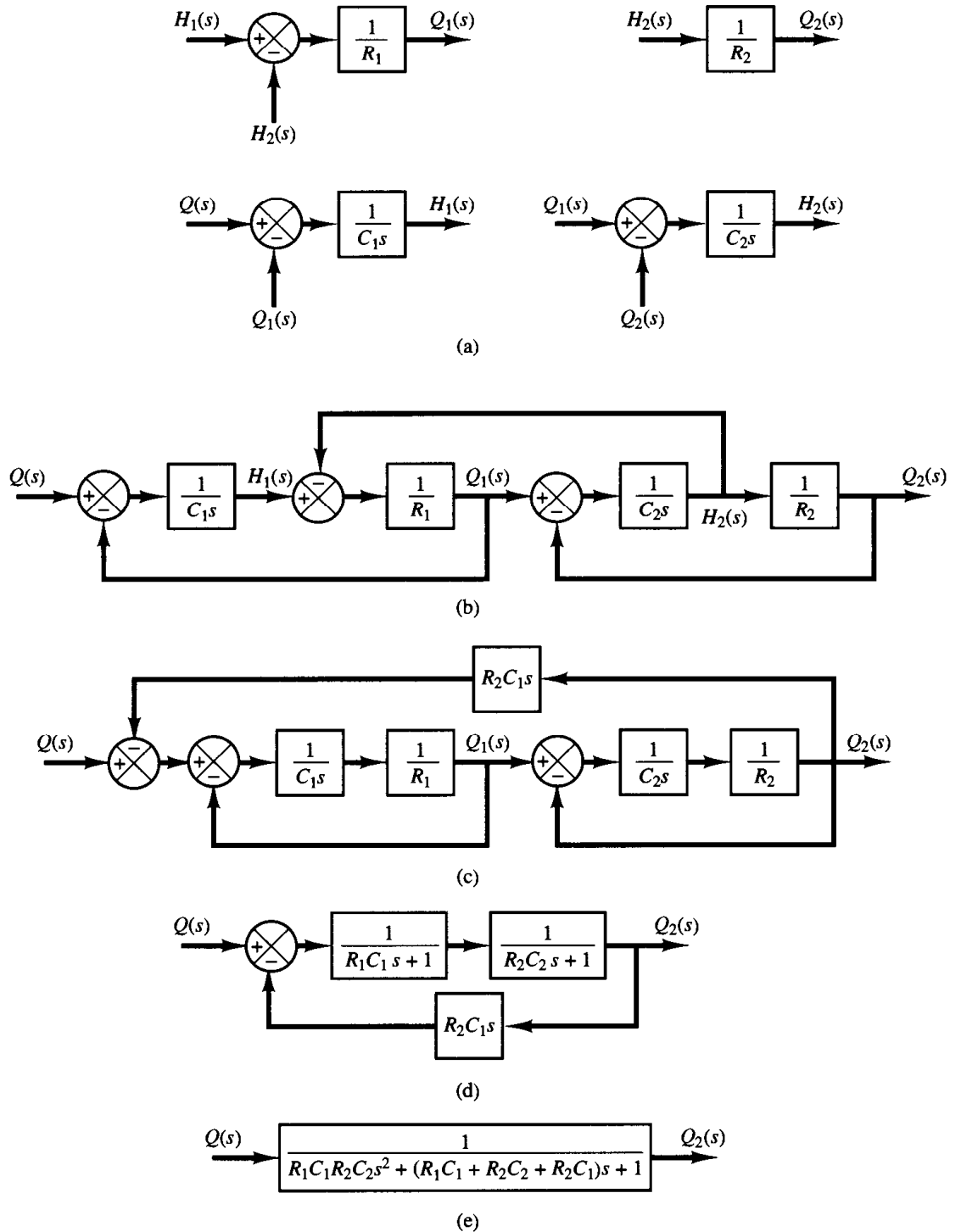


Figure 3-24
 (a) Elements of the block diagram of the system shown in Figure 3-23; (b) block diagram of the system; (c)–(e) successive reductions of the block diagram.

For conduction or convection heat transfer,

$$q = K \Delta\theta$$

where q = heat flow rate, kcal/sec
 $\Delta\theta$ = temperature difference, °C
 K = coefficient, kcal/sec °C

The coefficient K is given by

$$K = \frac{kA}{\Delta X}, \quad \text{for conduction}$$
$$= HA, \quad \text{for convection}$$

where k = thermal conductivity, kcal/m sec °C
 A = area normal to heat flow, m²
 ΔX = thickness of conductor, m
 H = convection coefficient, kcal/m² sec °C

Thermal resistance and thermal capacitance. The thermal resistance R for heat transfer between two substances may be defined as follows:

$$R = \frac{\text{change in temperature difference, } ^\circ\text{C}}{\text{change in heat flow rate, kcal/sec}}$$

The thermal resistance for conduction or convection heat transfer is given by

$$R = \frac{d(\Delta\theta)}{dq} = \frac{1}{K}$$

Since the thermal conductivity and convection coefficients are almost constant, the thermal resistance for either conduction or convection is constant.

The thermal capacitance C is defined by

$$C = \frac{\text{change in heat stored, kcal}}{\text{change in temperature, } ^\circ\text{C}}$$

or

$$C = mc$$

where m = mass of substance considered, kg
 c = specific heat of substance, kcal/kg °C

Thermal systems. Consider the system shown in Figure 3–25(a). It is assumed that the tank is insulated to eliminate heat loss to the surrounding air. It is also assumed that there is no heat storage in the insulation and that the liquid in the tank is perfectly mixed so that it is at a uniform temperature. Thus, a single temperature is used to describe the temperature of the liquid in the tank and of the outflowing liquid.

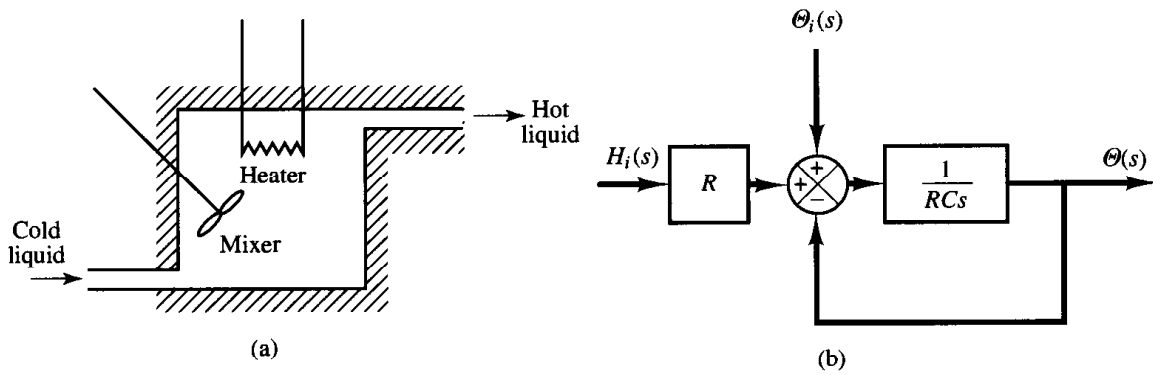


Figure 3-25
(a) Thermal system; (b) block diagram of the system.

Let us define

$\bar{\theta}_i$ = steady-state temperature of inflowing liquid, °C

$\bar{\theta}_o$ = steady-state temperature of outflowing liquid, °C

G = steady-state liquid flow rate, kg/sec

M = mass of liquid in tank, kg

c = specific heat of liquid, kcal/kg °C

R = thermal resistance, °C sec/kcal

C = thermal capacitance, kcal/°C

$\bar{H}$ = steady-state heat input rate, kcal/sec

Assume that the temperature of the inflowing liquid is kept constant and that the heat input rate to the system (heat supplied by the heater) is suddenly changed from $\bar{H}$ to $\bar{H} + h_i$, where h_i represents a small change in the heat input rate. The heat outflow rate will then change gradually from $\bar{H}$ to $\bar{H} + h_o$. The temperature of the outflowing liquid will also be changed from $\bar{\theta}_o$ to $\bar{\theta}_o + \theta$. For this case, h_o , C , and R are obtained, respectively, as

$$h_o = Gc\theta$$

$$C = Mc$$

$$R = \frac{\theta}{h_o} = \frac{1}{Gc}$$

The differential equation for this system is

$$C \frac{d\theta}{dt} = h_i - h_o$$

which may be rewritten as

$$RC \frac{d\theta}{dt} + \theta = Rh_i$$

Note that the time constant of the system is equal to RC or M/G seconds. The transfer function relating θ and h_i is given by

$$\frac{\Theta(s)}{H_i(s)} = \frac{R}{RCs + 1}$$

where $\Theta(s) = \mathcal{L}[\theta(t)]$ and $H_i(s) = \mathcal{L}[h_i(t)]$.

In practice, the temperature of the inflowing liquid may fluctuate and may act as a load disturbance. (If a constant outflow temperature is desired, an automatic controller may be installed to adjust the heat inflow rate to compensate for the fluctuations in the temperature of the inflowing liquid.) If the temperature of the inflowing liquid is suddenly changed from $\bar{\theta}_i$ to $\bar{\theta}_i + \theta_i$ while the heat input rate H and the liquid flow rate G are kept constant, then the heat outflow rate will be changed from $\bar{H}$ to $\bar{H} + h_o$, and the temperature of the outflowing liquid will be changed from $\bar{\theta}_o$ to $\bar{\theta}_o + \theta$. The differential equation for this case is

$$C \frac{d\theta}{dt} = Gc\theta_i - h_o$$

which may be rewritten

$$RC \frac{d\theta}{dt} + \theta = \theta_i$$

The transfer function relating θ and θ_i is given by

$$\frac{\Theta(s)}{\Theta_i(s)} = \frac{1}{RCs + 1}$$

where $\Theta(s) = \mathcal{L}[\theta(t)]$ and $\Theta_i(s) = \mathcal{L}[\theta_i(t)]$.

If the present thermal system is subjected to changes in both the temperature of the inflowing liquid and the heat input rate, while the liquid flow rate is kept constant, the change θ in the temperature of the outflowing liquid can be given by the following equation:

$$RC \frac{d\theta}{dt} + \theta = \theta_i + Rh_i$$

A block diagram corresponding to this case is shown in Figure 3–25(b). Notice that the system involves two inputs.

3–10 LINEARIZATION OF NONLINEAR MATHEMATICAL MODELS

In this section we present a linearization technique that is applicable to many nonlinear systems. The process of linearizing nonlinear systems is important, for by linearizing nonlinear equations, it is possible to apply numerous linear analysis methods that will produce information on the behavior of nonlinear systems. The linearization procedure presented here is based on the expansion of the nonlinear function into a

Taylor series about the operating point and the retention of only the linear term. Because we neglect higher-order terms of Taylor series expansion, these neglected terms must be small enough; that is, the variables deviate only slightly from the operating condition.

In what follows we shall first present mathematical aspects of the linearization technique and then apply the technique to a hydraulic servo system to obtain a linear model for the system.

Linear approximation of nonlinear mathematical models. To obtain a linear mathematical model for a nonlinear system, we assume that the variables deviate only slightly from some operating condition. Consider a system whose input is $x(t)$ and output is $y(t)$. The relationship between $y(t)$ and $x(t)$ is given by

$$y = f(x) \quad (3-75)$$

If the normal operating condition corresponds to $\bar{x}, \bar{y}$, then Equation (3-75) may be expanded into a Taylor series about this point as follows:

$$\begin{aligned} y &= f(x) \\ &= f(\bar{x}) + \frac{df}{dx}(x - \bar{x}) + \frac{1}{2!} \frac{d^2f}{dx^2}(x - \bar{x})^2 + \dots \end{aligned} \quad (3-76)$$

where the derivatives $df/dx, d^2f/dx^2, \dots$ are evaluated at $x = \bar{x}$. If the variation $x - \bar{x}$ is small, we may neglect the higher-order terms in $x - \bar{x}$. Then Equation (3-76) may be written as

$$y = \bar{y} + K(x - \bar{x}) \quad (3-77)$$

where

$$\begin{aligned} \bar{y} &= f(\bar{x}) \\ K &= \left. \frac{df}{dx} \right|_{x=\bar{x}} \end{aligned}$$

Equation (3-77) may be rewritten as

$$y - \bar{y} = K(x - \bar{x}) \quad (3-78)$$

which indicates that $y - \bar{y}$ is proportional to $x - \bar{x}$. Equation (3-78) gives a linear mathematical model for the nonlinear system given by Equation (3-75) near the operating point $x = \bar{x}, y = \bar{y}$.

Next, consider a nonlinear system whose output y is a function of two inputs x_1 and x_2 , so that

$$y = f(x_1, x_2) \quad (3-79)$$

To obtain a linear approximation to this nonlinear system, we may expand Equation (3-79) into a Taylor series about the normal operating point $\bar{x}_1, \bar{x}_2$. Then Equation (3-79) becomes

$$\begin{aligned}
y = f(\bar{x}_1, \bar{x}_2) &+ \left[\frac{\partial f}{\partial x_1} (x_1 - \bar{x}_1) + \frac{\partial f}{\partial x_2} (x_2 - \bar{x}_2) \right] \\
&+ \frac{1}{2!} \left[\frac{\partial^2 f}{\partial x_1^2} (x_1 - \bar{x}_1)^2 + 2 \frac{\partial^2 f}{\partial x_1 \partial x_2} (x_1 - \bar{x}_1)(x_2 - \bar{x}_2) \right. \\
&\left. + \frac{\partial^2 f}{\partial x_2^2} (x_2 - \bar{x}_2)^2 \right] + \dots
\end{aligned}$$

where the partial derivatives are evaluated at $x_1 = \bar{x}_1, x_2 = \bar{x}_2$. Near the normal operating point, the higher-order terms may be neglected. The linear mathematical model of this nonlinear system in the neighborhood of the normal operating condition is then given by

$$y - \bar{y} = K_1(x_1 - \bar{x}_1) + K_2(x_2 - \bar{x}_2)$$

where

$$\bar{y} = f(\bar{x}_1, \bar{x}_2)$$

$$K_1 = \left. \frac{\partial f}{\partial x_1} \right|_{x_1=\bar{x}_1, x_2=\bar{x}_2}$$

$$K_2 = \left. \frac{\partial f}{\partial x_2} \right|_{x_1=\bar{x}_1, x_2=\bar{x}_2}$$

The linearization technique presented here is valid in the vicinity of the operating condition. If the operating conditions vary widely, however, such linearized equations are not adequate, and nonlinear equations must be dealt with. It is important to remember that a particular mathematical model used in analysis and design may accurately represent the dynamics of an actual system for certain operating conditions, but may not be accurate for other operating conditions.

Linearization of a hydraulic servo system. Figure 3-26(a) shows a hydraulic servomotor. It is essentially a pilot-valve-controlled hydraulic power amplifier and actuator. The pilot valve is a balanced valve, in the sense that the pressure forces acting on it are all balanced. A very large power output can be controlled by a pilot valve, which can be positioned with very little power.

In practice, the ports shown in Figure 3-26(a) are often made wider than the corresponding valves. In such a case, there is always leakage through the valves. Such leakage improves both the sensitivity and the linearity of the hydraulic servomotor. In the following analysis we shall make the assumption that the ports are made wider than the valves, that is, the valves are underlapped. [Note that sometimes a dither signal, a high-frequency signal of very small amplitude (with respect to the maximum displacement of the valve), is superimposed on the motion of the pilot valve. This also improves the sensitivity and linearity. In this case also there is leakage through the valve.]

We shall apply the linearization technique just presented to obtain a linearized mathematical model of the hydraulic servomotor. We assume that the valve is underlapped and symmetrical and admits hydraulic fluid under high pressure into a power cylinder that contains a large piston, so that a large hydraulic force is established to

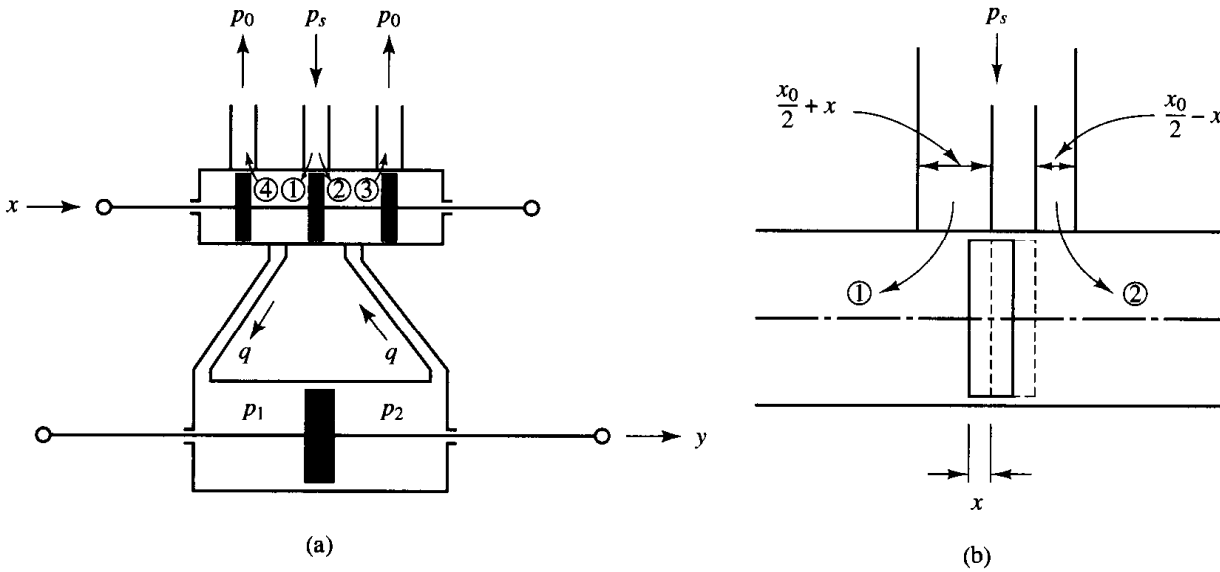


Figure 3-26

(a) Hydraulic servo system; (b) enlarged diagram of the valve orifice area.

move a load. We assume that the load inertia and friction are small compared to the large hydraulic force. In the present analysis, the hydraulic fluid is assumed to be incompressible and the inertia force of the power piston negligible. We also assume that, as is usually the case, the orifice area (the width of the slot in the valve sleeve) at each port is proportional to the valve displacement x .

In Figure 3-26(b) we have an enlarged diagram of the valve orifice area. Let us define the valve orifice areas of ports 1, 2, 3, 4 as A_1, A_2, A_3, A_4 , respectively. Also, define the flow rates through ports 1, 2, 3, 4 as q_1, q_2, q_3, q_4 , respectively. Note that, since the valve is symmetrical, $A_1 = A_3$ and $A_2 = A_4$. Assuming the displacement x to be small, we obtain

$$A_1 = A_3 = k \left(\frac{x_0}{2} + x \right)$$

$$A_2 = A_4 = k \left(\frac{x_0}{2} - x \right)$$

where k is a constant.

Furthermore, we shall assume that the return pressure p_0 in the return line is small and thus can be neglected. Then, referring to Figure 3-26(a), flow rates through valve orifices are

$$q_1 = c_1 A_1 \sqrt{\frac{2g}{\gamma} (p_s - p_1)} = C_1 \sqrt{p_s - p_1} \left(\frac{x_0}{2} + x \right)$$

$$q_2 = c_2 A_2 \sqrt{\frac{2g}{\gamma} (p_s - p_2)} = C_2 \sqrt{p_s - p_2} \left(\frac{x_0}{2} - x \right)$$

$$q_3 = c_1 A_3 \sqrt{\frac{2g}{\gamma} (p_2 - p_0)} = C_1 \sqrt{p_2 - p_0} \left(\frac{x_0}{2} + x \right) = C_1 \sqrt{p_2} \left(\frac{x_0}{2} + x \right)$$

$$q_4 = c_2 A_4 \sqrt{\frac{2g}{\gamma} (p_1 - p_0)} = C_2 \sqrt{p_1 - p_0} \left(\frac{x_0}{2} - x \right) = C_2 \sqrt{p_1} \left(\frac{x_0}{2} - x \right)$$

where $C_1 = c_1 k \sqrt{2g/\gamma}$ and $C_2 = c_2 k \sqrt{2g/\gamma}$, and γ is the specific weight and is given by $\gamma = \rho g$, where ρ is mass density and g is the acceleration of gravity. The flow rate q to the left-hand side of the power piston is

$$q = q_1 - q_4 = C_1 \sqrt{p_s - p_1} \left(\frac{x_0}{2} + x \right) - C_2 \sqrt{p_1} \left(\frac{x_0}{2} - x \right) \quad (3-80)$$

The flow rate from the right-hand side of the power piston to the drain is the same as this q and is given by

$$q = q_3 - q_2 = C_1 \sqrt{p_2} \left(\frac{x_0}{2} + x \right) - C_2 \sqrt{p_s - p_2} \left(\frac{x_0}{2} - x \right)$$

Note that the fluid is incompressible and that the valve is symmetrical. So we have $q_1 = q_3$ and $q_2 = q_4$. By equating q_1 and q_3 , we obtain

$$p_s - p_1 = p_2$$

or

$$p_s = p_1 + p_2$$

If we define the pressure difference across the power piston as Δp or

$$\Delta p = p_1 - p_2$$

then

$$p_1 = \frac{p_s + \Delta p}{2}, \quad p_2 = \frac{p_s - \Delta p}{2}$$

For the symmetrical valve shown in Figure 3-26(a), the pressure in each side of the power piston is $\frac{1}{2}p_s$ when no load is applied, or $\Delta p = 0$. As the spool valve is displaced, the pressure in one line increases as the pressure in the other line decreases by the same amount.

In terms of p_s and Δp , we can rewrite the flow rate q given by Equation (3-80) as

$$q = q_1 - q_4 = C_1 \sqrt{\frac{p_s - \Delta p}{2}} \left(\frac{x_0}{2} + x \right) - C_2 \sqrt{\frac{p_s + \Delta p}{2}} \left(\frac{x_0}{2} - x \right)$$

Noting that the supply pressure p_s is constant, the flow rate q can be written as a function of the valve displacement x and pressure difference Δp , or

$$q = C_1 \sqrt{\frac{p_s - \Delta p}{2}} \left(\frac{x_0}{2} + x \right) - C_2 \sqrt{\frac{p_s + \Delta p}{2}} \left(\frac{x_0}{2} - x \right) = f(x, \Delta p)$$

By applying the linearization technique presented earlier in this section to this case, the linearized equation about point $x = \bar{x}$, $\Delta p = \Delta \bar{p}$, $q = \bar{q}$ is

$$q - \bar{q} = a(x - \bar{x}) + b(\Delta p - \Delta \bar{p}) \quad (3-81)$$

where

$$\bar{q} = f(\bar{x}, \Delta \bar{p})$$

$$a = \left. \frac{\partial f}{\partial x} \right|_{x=\bar{x}, \Delta p=\Delta \bar{p}} = C_1 \sqrt{\frac{p_s - \Delta \bar{p}}{2}} + C_2 \sqrt{\frac{p_s + \Delta \bar{p}}{2}}$$

$$b = \left. \frac{\partial f}{\partial \Delta p} \right|_{x=\bar{x}, \Delta p=\Delta \bar{p}} = - \left[\frac{C_1}{2\sqrt{2} \sqrt{p_s - \Delta \bar{p}}} \left(\frac{x_0}{2} + \bar{x} \right) + \frac{C_2}{2\sqrt{2} \sqrt{p_s + \Delta \bar{p}}} \left(\frac{x_0}{2} - \bar{x} \right) \right] < 0$$

Coefficients a and b here are called *valve coefficients*. Equation (3-81) is a linearized mathematical model of the spool valve near an operating point $x = \bar{x}$, $\Delta p = \Delta \bar{p}$, $q = \bar{q}$. The values of valve coefficients a and b vary with the operating point. Note that $\partial f / \partial \Delta p$ is negative and so b is negative.

Since the normal operating point is the point where $\bar{x} = 0$, $\Delta \bar{p} = 0$, $\bar{q} = 0$, near the normal operating point, Equation (3-81) becomes

$$q = K_1 x - K_2 \Delta p \quad (3-82)$$

where

$$K_1 = (C_1 + C_2) \sqrt{\frac{p_s}{2}} > 0$$

$$K_2 = (C_1 + C_2) \frac{x_0}{4\sqrt{2} \sqrt{p_s}} > 0$$

Equation (3-82) is a linearized mathematical model of the spool valve near the origin ($\bar{x} = 0$, $\Delta \bar{p} = 0$, $\bar{q} = 0$). Note that the region near the origin is most important in this kind of system, because the system operation usually occurs near this point. (For a derivation of a mathematical model of a hydraulic servo system when the load reactive forces are not negligible, see Problem A-3-20.)

EXAMPLE PROBLEMS AND SOLUTIONS

A-3-1. Simplify the block diagram shown in Figure 3-27.

Solution. First, move the branch point of the path involving H_1 outside the loop involving H_2 , as shown in Figure 3-28(a). Then eliminating two loops results in Figure 3-28(b). Combining two blocks into one gives Figure 3-28(c).

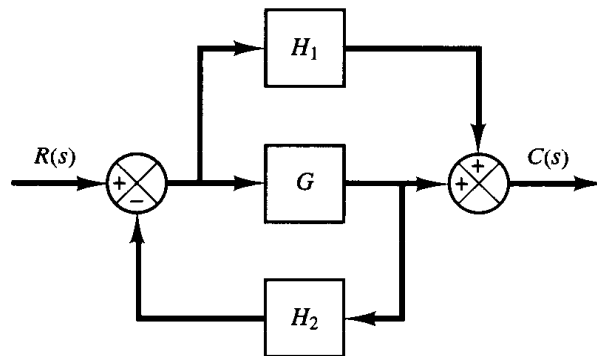


Figure 3-27
Block diagram of a system.

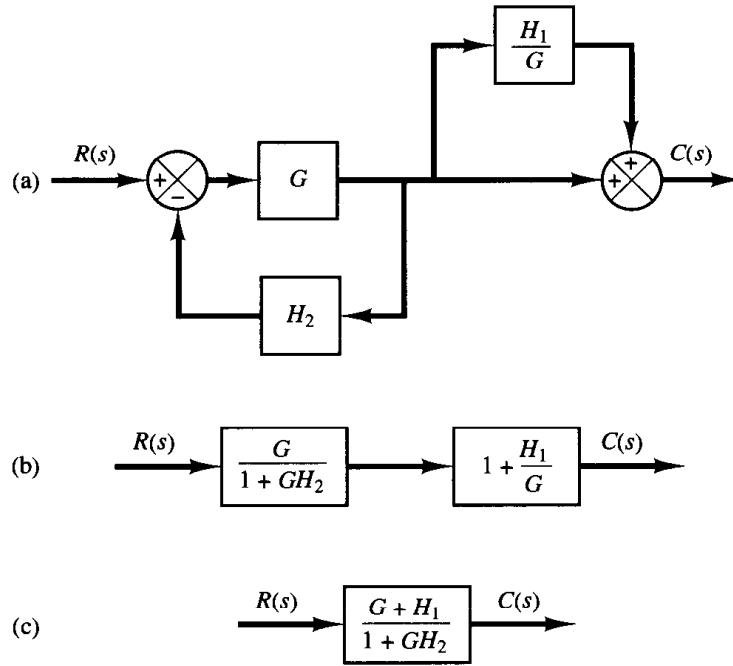


Figure 3-28
Simplified block
diagrams for the
system shown in
Figure 3-27.

A-3-2. Simplify the block diagram shown in Figure 3-29. Obtain the transfer function relating $C(s)$ and $R(s)$.

Solution. The block diagram of Figure 3-29 can be modified to that shown in Figure 3-30(a). Eliminating the minor feedforward path, we obtain Figure 3-30(b), which can be simplified to that shown in Figure 3-30(c). The transfer function $C(s)/R(s)$ is thus given by

$$\frac{C(s)}{R(s)} = G_1 G_2 + G_2 + 1$$

The same result can also be obtained by proceeding as follows: Since signal $X(s)$ is the sum of two signals $G_1 R(s)$ and $R(s)$, we have

$$X(s) = G_1 R(s) + R(s)$$

The output signal $C(s)$ is the sum of $G_2 X(s)$ and $R(s)$. Hence

$$C(s) = G_2 X(s) + R(s) = G_2 [G_1 R(s) + R(s)] + R(s)$$

And so we have the same result as before:

$$\frac{C(s)}{R(s)} = G_1 G_2 + G_2 + 1$$

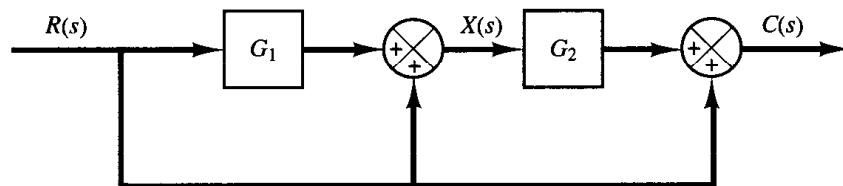


Figure 3-29
Block diagram of a
system.

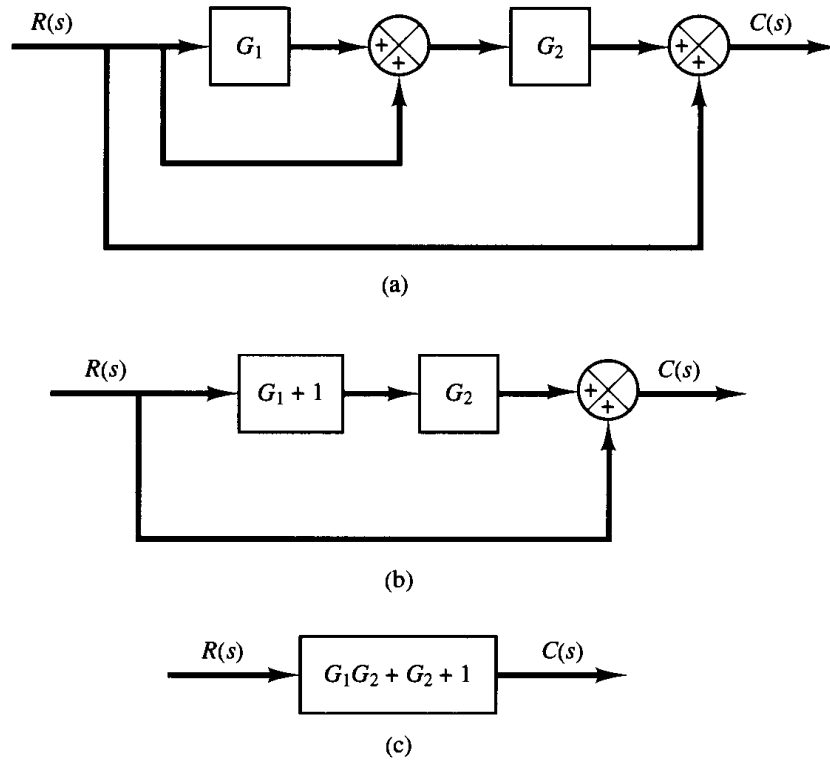


Figure 3-30
Reduction of the
block diagram shown
in Figure 3-29.

A-3-3. Show that for the differential equation system

$$\ddot{y} + a_1\ddot{y} + a_2\dot{y} + a_3y = b_0\ddot{u} + b_1\dot{u} + b_2\dot{u} + b_3u \quad (3-83)$$

state and output equations can be given, respectively, by

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_3 & -a_2 & -a_1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix} u \quad (3-84)$$

and

$$y = [1 \quad 0 \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \beta_0 u \quad (3-85)$$

where state variables are defined by

$$x_1 = y - \beta_0 u$$

$$x_2 = \dot{y} - \beta_0 \dot{u} - \beta_1 u = \dot{x}_1 - \beta_1 u$$

$$x_3 = \ddot{y} - \beta_0 \ddot{u} - \beta_1 \dot{u} - \beta_2 u = \dot{x}_2 - \beta_2 u$$

and

$$\beta_0 = b_0$$

$$\beta_1 = b_1 - a_1\beta_0$$

$$\beta_2 = b_2 - a_1\beta_1 - a_2\beta_0$$

$$\beta_3 = b_3 - a_1\beta_2 - a_2\beta_1 - a_3\beta_0$$

Solution. From the definition of state variables x_2 and x_3 , we have

$$\dot{x}_1 = x_2 + \beta_1 u \quad (3-86)$$

$$\dot{x}_2 = x_3 + \beta_2 u \quad (3-87)$$

To derive the equation for $\dot{x}_3$, we first note from Equation (3-83) that

$$\ddot{y} = -a_1 \ddot{y} - a_2 \dot{y} - a_3 y + b_0 \ddot{u} + b_1 \dot{u} + b_2 \dot{u} + b_3 u$$

Since

$$x_3 = \ddot{y} - \beta_0 \ddot{u} - \beta_1 \dot{u} - \beta_2 u$$

we have

$$\begin{aligned} \dot{x}_3 &= \ddot{y} - \beta_0 \ddot{u} - \beta_1 \dot{u} - \beta_2 \dot{u} \\ &= (-a_1 \ddot{y} - a_2 \dot{y} - a_3 y) + b_0 \ddot{u} + b_1 \dot{u} + b_2 \dot{u} + b_3 u - \beta_0 \ddot{u} - \beta_1 \dot{u} - \beta_2 \dot{u} \\ &= -a_1 (\ddot{y} - \beta_0 \ddot{u} - \beta_1 \dot{u} - \beta_2 u) - a_1 \beta_0 \ddot{u} - a_1 \beta_1 \dot{u} - a_1 \beta_2 u \\ &\quad - a_2 (\dot{y} - \beta_0 \dot{u} - \beta_1 u) - a_2 \beta_0 \dot{u} - a_2 \beta_1 u - a_3 (y - \beta_0 u) - a_3 \beta_0 u \\ &\quad + b_0 \ddot{u} + b_1 \dot{u} + b_2 \dot{u} + b_3 u - \beta_0 \ddot{u} - \beta_1 \dot{u} - \beta_2 \dot{u} \\ &= -a_1 x_3 - a_2 x_2 - a_3 x_1 + (b_0 - \beta_0) \ddot{u} + (b_1 - \beta_1 - a_1 \beta_0) \dot{u} \\ &\quad + (b_2 - \beta_2 - a_1 \beta_1 - a_2 \beta_0) \dot{u} + (b_3 - a_1 \beta_2 - a_2 \beta_1 - a_3 \beta_0) u \\ &= -a_1 x_3 - a_2 x_2 - a_3 x_1 + (b_3 - a_1 \beta_2 - a_2 \beta_1 - a_3 \beta_0) u \\ &= -a_1 x_3 - a_2 x_2 - a_3 x_1 + \beta_3 u \end{aligned}$$

Hence, we get

$$\dot{x}_3 = -a_3 x_1 - a_2 x_2 - a_1 x_3 + \beta_3 u \quad (3-88)$$

Combining Equations (3-86), (3-87), and (3-88) into a vector-matrix differential equation, we obtain Equation (3-84). Also, from the definition of state variable x_1 , we get the output equation given by Equation (3-85).

A-3-4 Obtain a state-space model of the system shown in Figure 3-31.

Solution. The system involves one integrator and two delayed integrators. The output of each integrator or delayed integrator can be a state variable. Let us define the output of the plant as x_1 , the output of the controller as x_2 , and the output of the sensor as x_3 . Then we obtain

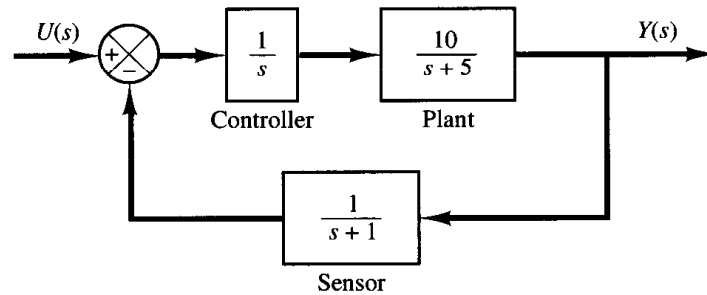


Figure 3-31
Control system.

$$\begin{aligned}\frac{X_1(s)}{X_2(s)} &= \frac{10}{s+5} \\ \frac{X_2(s)}{U(s) - X_3(s)} &= \frac{1}{s} \\ \frac{X_3(s)}{X_1(s)} &= \frac{1}{s+1} \\ Y(s) &= X_1(s)\end{aligned}$$

which can be rewritten as

$$\begin{aligned}sX_1(s) &= -5X_1(s) + 10X_2(s) \\ sX_2(s) &= -X_3(s) + U(s) \\ sX_3(s) &= X_1(s) - X_3(s) \\ Y(s) &= X_1(s)\end{aligned}$$

By taking the inverse Laplace transforms of the preceding four equations, we obtain

$$\begin{aligned}\dot{x}_1 &= -5x_1 + 10x_2 \\ \dot{x}_2 &= -x_3 + u \\ \dot{x}_3 &= x_1 - x_3 \\ y &= x_1\end{aligned}$$

Thus, a state-space model of the system in the standard form is given by

$$\begin{aligned}\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} &= \begin{bmatrix} -5 & 10 & 0 \\ 0 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} u \\ y &= [1 \quad 0 \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\end{aligned}$$

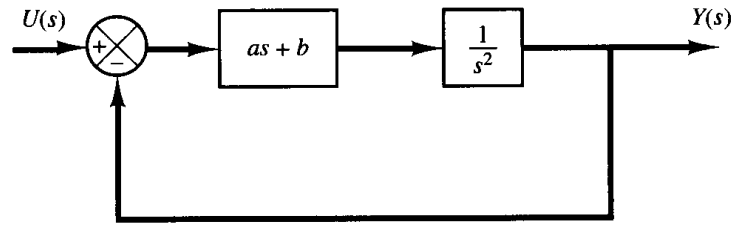
It is important to note that this is not the only state-space representation of the system. Many other state-space representations are possible. However, the number of state variables is the same in any state-space representation of the same system. In the present system, the number of state variables is three, regardless of what variables are chosen as state variables.

A-3-5. Obtain a state-space model for the system shown in Figure 3–32(a).

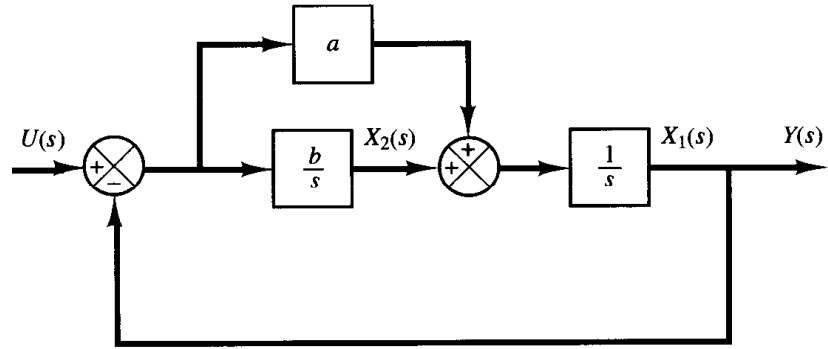
Solution. First, notice that $(as + b)/s^2$ involves a derivative term. Such a derivative term may be avoided if we modify $(as + b)/s^2$ as

$$\frac{as + b}{s^2} = \left(a + \frac{b}{s}\right) \frac{1}{s}$$

Using this modification, the block diagram of Figure 3–32(a) can be modified to that shown in Figure 3–32(b).



(a)



(b)

Figure 3–32

(a) Control system;
(b) modified block diagram.

Define the outputs of the integrators as state variables, as shown in Figure 3–32(b). Then from Figure 3–32(b) we obtain

$$\frac{X_1(s)}{X_2(s) + a[U(s) - X_1(s)]} = \frac{1}{s}$$

$$\frac{X_2(s)}{U(s) - X_1(s)} = \frac{b}{s}$$

$$Y(s) = X_1(s)$$

which may be modified to

$$sX_1(s) = X_2(s) + a[U(s) - X_1(s)]$$

$$sX_2(s) = -bX_1(s) + bU(s)$$

$$Y(s) = X_1(s)$$

Taking the inverse Laplace transforms of the preceding three equations, we obtain

$$\dot{x}_1 = -ax_1 + x_2 + au$$

$$\dot{x}_2 = -bx_1 + bu$$

$$y = x_1$$

Rewriting the state and output equations in the standard vector-matrix form, we obtain

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -a & 1 \\ -b & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} a \\ b \end{bmatrix} u$$

$$y = [1 \quad 0] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

A-3-6. Obtain a state-space representation of the system shown in Figure 3-33(a).

Solution. In this problem, first expand $(s + z)/(s + p)$ into partial fractions.

$$\frac{s + z}{s + p} = 1 + \frac{z - p}{s + p}$$

Next convert $K/[s(s + a)]$ into the product of K/s and $1/(s + a)$. Then redraw the block diagram, as shown in Figure 3-33(b). Defining a set of state variables, as shown in Figure 3-33(b), we obtain the following equations:

$$\dot{x}_1 = -ax_1 + x_2$$

$$\dot{x}_2 = -Kx_1 + Kx_3 + Ku$$

$$\dot{x}_3 = -(z - p)x_1 - px_3 + (z - p)u$$

$$y = x_1$$

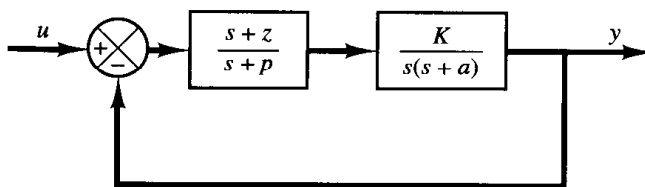
Rewriting gives

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -a & 1 & 0 \\ -K & 0 & K \\ -(z - p) & 0 & -p \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ K \\ z - p \end{bmatrix} u$$

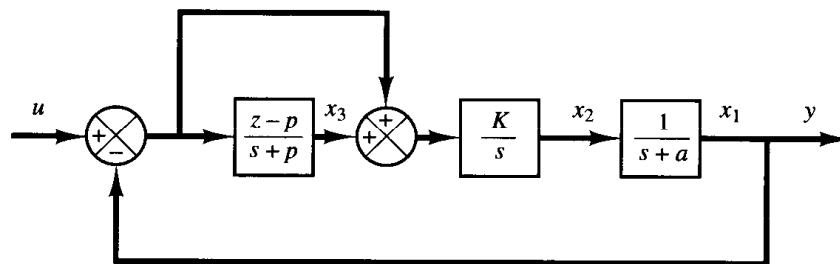
$$y = [1 \quad 0 \quad 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Notice that the output of the integrator and the outputs of the first-order delayed integrators $[1/(s + a)$ and $(z - p)/(s + p)]$ are chosen as state variables. It is important to remember that the output of the block $(s + z)/(s + p)$ in Figure 3-33(a) cannot be a state variable, because this block involves a derivative term, $s + z$.

A-3-7. Gyros for sensing angular motion are commonly used in inertial guidance systems, autopilot systems, and the like. Figure 3-34(a) shows a single-degree-of-freedom gyro. The spinning wheel is



(a)



(b)

Figure 3-33

(a) Control system;
(b) block diagram
defining state vari-
ables for the system.

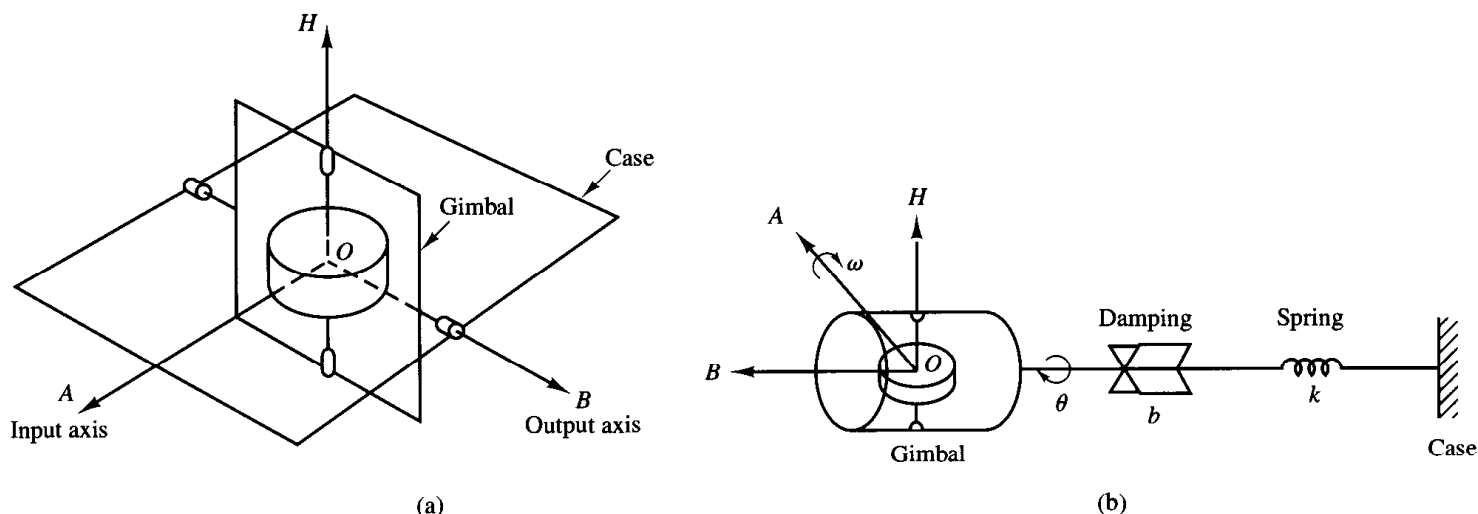


Figure 3-34

(a) Schematic diagram of a single-degree-of-freedom gyro; (b) functional diagram of the gyro shown in part (a).

mounted in a movable gimbal, which is, in turn, mounted in a gyro case. The gimbal is free to move relative to the case about the output axis OB . Note that the output axis is perpendicular to the wheel spin axis. The input axis around which a turning rate, or angle, is measured is perpendicular to both the output and spin axes. Information on the input signal (the turning rate or angle around the input axis) is obtained from the resulting motion of the gimbal about the output axis, relative to the case.

Figure 3-34(b) shows a functional diagram of the gyro system. The equation of motion about the output axis can be obtained by equating the rate of change of angular momentum to the sum of the external torques.

The change in angular momentum about axis OB consists of two parts: $I\ddot{\theta}$, the change due to acceleration of the gimbal around axis OB , and $-H\omega \cos \theta$, the change due to the turning of the wheel angular-momentum vector around axis OA . The external torques consist of $-b\dot{\theta}$, the damping torque, and $-k\theta$, the spring torque. Thus the equation of the gyro system is

$$I\ddot{\theta} - H\omega \cos \theta = -b\dot{\theta} - k\theta$$

or

$$I\ddot{\theta} + b\dot{\theta} + k\theta = H\omega \cos \theta \quad (3-89)$$

In practice, θ is a very small angle, usually not more than $\pm 2.5^\circ$.

Obtain a state-space representation of the gyro system.

Solution. In this system, θ and $\dot{\theta}$ may be chosen as state variables. The input variable is ω and the output variable is θ . Let us define

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix}, \quad u = \omega, \quad y = \theta$$

Then Equation (3-89) can be written as follows:

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -\frac{k}{I}x_1 - \frac{b}{I}x_2 + \frac{H}{I}u \cos x_1 \end{aligned}$$

or

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, u)$$

where

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad \mathbf{f}(\mathbf{x}, u) = \begin{bmatrix} f_1(\mathbf{x}, u) \\ f_2(\mathbf{x}, u) \end{bmatrix} = \begin{bmatrix} x_2 \\ -\frac{k}{I}x_1 - \frac{b}{I}x_2 + \frac{H}{I}u \cos x_1 \end{bmatrix}$$

Clearly, $f_2(\mathbf{x}, u)$ involves a nonlinear term in x_1 and u . By expanding $\cos x_1$ into its series representation,

$$\cos x_1 = 1 - \frac{1}{2}x_1^2 + \cdots$$

and noting that x_1 is a very small angle, we may approximate $\cos x_1$ by unity to obtain the following linearized state equation:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\frac{k}{I} & -\frac{b}{I} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{H}{I} \end{bmatrix} u$$

The output equation is

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

A-3-8. Consider a system defined by the following state-space equations:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -5 & -1 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 2 \\ 5 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Obtain the transfer function $G(s)$ of the system.

Solution. Referring to Equation (3-32), the transfer function of the system can be obtained as follows (note that $D = 0$ in this case):

$$\begin{aligned} G(s) &= \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} \\ &= \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} s+5 & 1 \\ -3 & s+1 \end{bmatrix}^{-1} \begin{bmatrix} 2 \\ 5 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} \frac{s+1}{(s+2)(s+4)} & \frac{-1}{(s+2)(s+4)} \\ \frac{3}{(s+2)(s+4)} & \frac{s+5}{(s+2)(s+4)} \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix} \\ &= \frac{12s+59}{(s+2)(s+4)} \end{aligned}$$

A-3-9. Figure 3-35(a) shows a schematic diagram of an automobile suspension system. As the car moves along the road, the vertical displacements at the tires act as the motion excitation to the automobile suspension system. The motion of this system consists of a translational motion of the center of mass and a rotational motion about the center of mass. Mathematical modeling of the complete system is quite complicated.

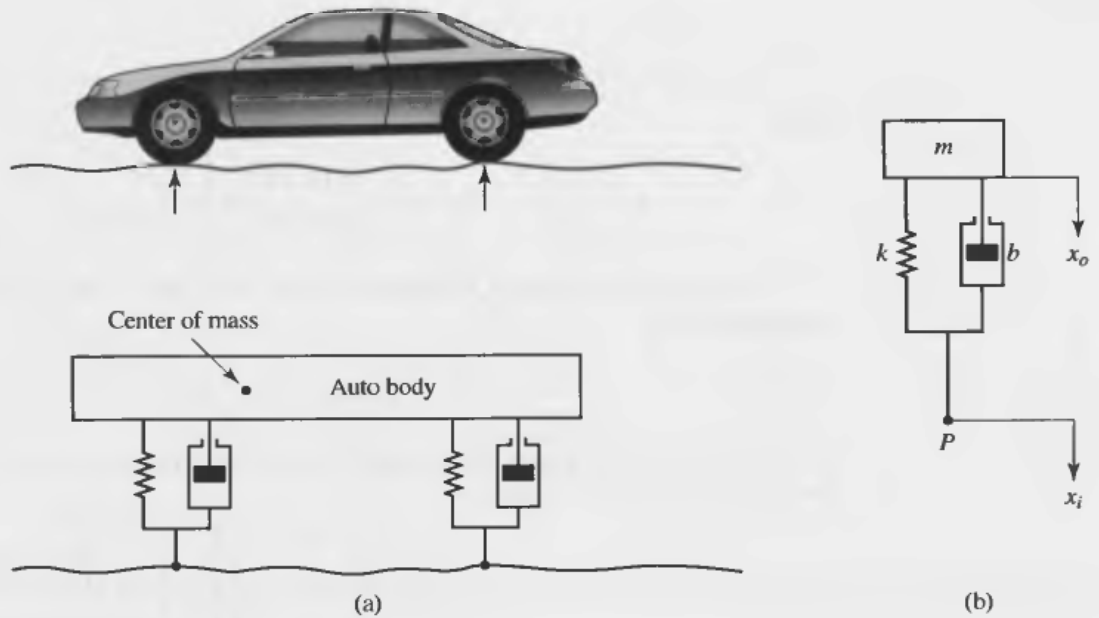


Figure 3-35
(a) Automobile suspension system;
(b) simplified suspension system.

A very simplified version of the suspension system is shown in Figure 3-35(b). Assuming that the motion x_i at point P is the input to the system and the vertical motion x_o of the body is the output, obtain the transfer function $X_o(s)/X_i(s)$. (Consider the motion of the body only in the vertical direction.) Displacement x_o is measured from the equilibrium position in the absence of input x_i .

Solution. The equation of motion for the system shown in Figure 3-35(b) is

$$m\ddot{x}_o + b(\dot{x}_o - \dot{x}_i) + k(x_o - x_i) = 0$$

or

$$m\ddot{x}_o + b\dot{x}_o + kx_o = b\dot{x}_i + kx_i$$

Taking the Laplace transform of this last equation, assuming zero initial conditions, we obtain

$$(ms^2 + bs + k)X_o(s) = (bs + k)X_i(s)$$

Hence the transfer function $X_o(s)/X_i(s)$ is given by

$$\frac{X_o(s)}{X_i(s)} = \frac{bs + k}{ms^2 + bs + k}$$

A-3-10. Obtain the transfer function $Y(s)/U(s)$ of the system shown in Figure 3-36. (Similar to the system of Problem A-3-9, this is also a simplified version of an automobile or motorcycle suspension system.)

Solution. Applying the Newton's second law to the system, we obtain

$$m_1\ddot{x} = k_2(y - x) + b(\dot{y} - \dot{x}) + k_1(u - x)$$

$$m_2\ddot{y} = -k_2(y - x) - b(\dot{y} - \dot{x})$$

Hence, we have

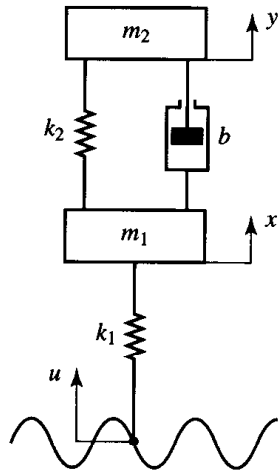


Figure 3-36
Suspension system.

$$\begin{aligned} m_1 \ddot{x} + b \dot{x} + (k_1 + k_2)x &= b \dot{y} + k_2 y + k_1 u \\ m_2 \ddot{y} + b \dot{y} + k_2 y &= b \dot{x} + k_2 x \end{aligned}$$

Taking Laplace transforms of these two equations, assuming zero initial conditions, we obtain

$$\begin{aligned} [m_1 s^2 + bs + (k_1 + k_2)]X(s) &= (bs + k_2)Y(s) + k_1 U(s) \\ [m_2 s^2 + bs + k_2]Y(s) &= (bs + k_2)X(s) \end{aligned}$$

Eliminating $X(s)$ from the last two equations, we have

$$(m_1 s^2 + bs + k_1 + k_2) \frac{m_2 s^2 + bs + k_2}{bs + k_2} Y(s) = (bs + k_2)Y(s) + k_1 U(s)$$

which yields

$$\frac{Y(s)}{U(s)} = \frac{k_1(bs + k_2)}{m_1 m_2 s^4 + (m_1 + m_2)bs^3 + [k_1 m_2 + (m_1 + m_2)k_2]s^2 + k_1 bs + k_1 k_2}$$

- A-3-11.** Consider the electrical circuit shown in Figure 3-37. Obtain the transfer function $E_o(s)/E_i(s)$ by use of the block diagram approach.

Solution. Equations for the circuits are

$$\frac{1}{C_1} \int (i_1 - i_2) dt + R_1 i_1 = e_i \quad (3-90)$$

$$\frac{1}{C_1} \int (i_2 - i_1) dt + R_2 i_2 + \frac{1}{C_2} \int i_2 dt = 0 \quad (3-91)$$

$$\frac{1}{C_2} \int i_2 dt = e_o \quad (3-92)$$

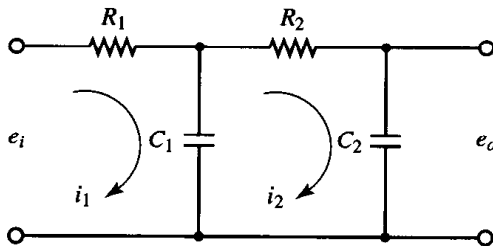


Figure 3-37
Electrical circuit.

The Laplace transforms of Equations (3–90), (3–91), and (3–92), with zero initial conditions, give

$$\frac{1}{C_1 s} [I_1(s) - I_2(s)] + R_1 I_1(s) = E_i(s) \quad (3-93)$$

$$\frac{1}{C_1 s} [I_2(s) - I_1(s)] + R_2 I_2(s) + \frac{1}{C_2 s} I_2(s) = 0 \quad (3-94)$$

$$\frac{1}{C_2 s} I_2(s) = E_o(s) \quad (3-95)$$

Equation (3–93) can be rewritten as

$$C_1 s [E_i(s) - R_1 I_1(s)] = I_1(s) - I_2(s) \quad (3-96)$$

Equation (3–96) gives the block diagram shown in Figure 3–38(a). Equation (3–94) can be modified to

$$I_2(s) = \frac{C_2 s}{R_2 C_2 s + 1} \frac{1}{C_1 s} [I_1(s) - I_2(s)] \quad (3-97)$$

Equation (3–97) yields the block diagram shown in Figure 3–38(b). Also, Equation (3–95) gives the block diagram shown in Figure 3–38(c). Combining the block diagrams of Figures 3–38(a), (b), and (c), we obtain Figure 3–39(a). This block diagram can be successively modified as shown in Figures 3–39(b) through (e). Thus we obtained the transfer function $E_o(s)/E_i(s)$ of the system. [This is the same as that we derived earlier for the same electrical circuit. See Equation (3–66).]

- A-3-12.** Obtain the transfer function of the mechanical system shown in Figure 3–40(a). Also obtain the transfer function of the electrical system shown in Figure 3–40(b). Show that the transfer functions of the two systems are of identical form and thus they are analogous systems.

Solution. The equations of motion for the mechanical system shown in Figure 3–40(a) are

$$b_1(\dot{x}_i - \dot{x}_o) + k_1(x_i - x_o) = b_2(\dot{x}_o - \dot{y})$$

$$b_2(\dot{x}_o - \dot{y}) = k_2 y$$

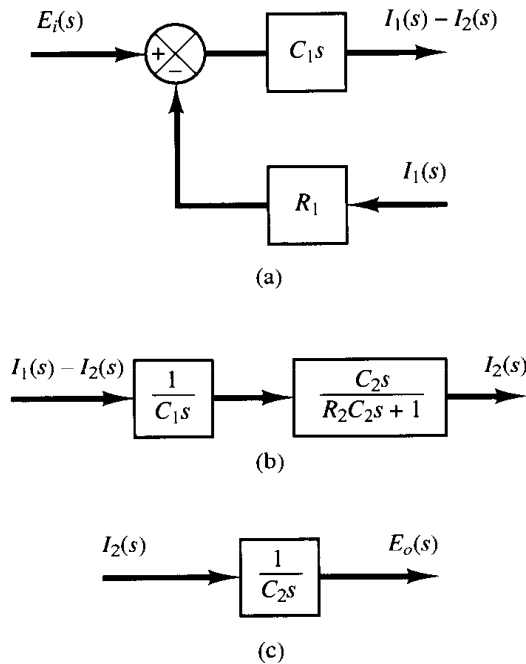


Figure 3–38

Block diagrams: (a) corresponding to Equation (3–96); (b) corresponding to Equation (3–97); (c) corresponding to Equation (3–95).

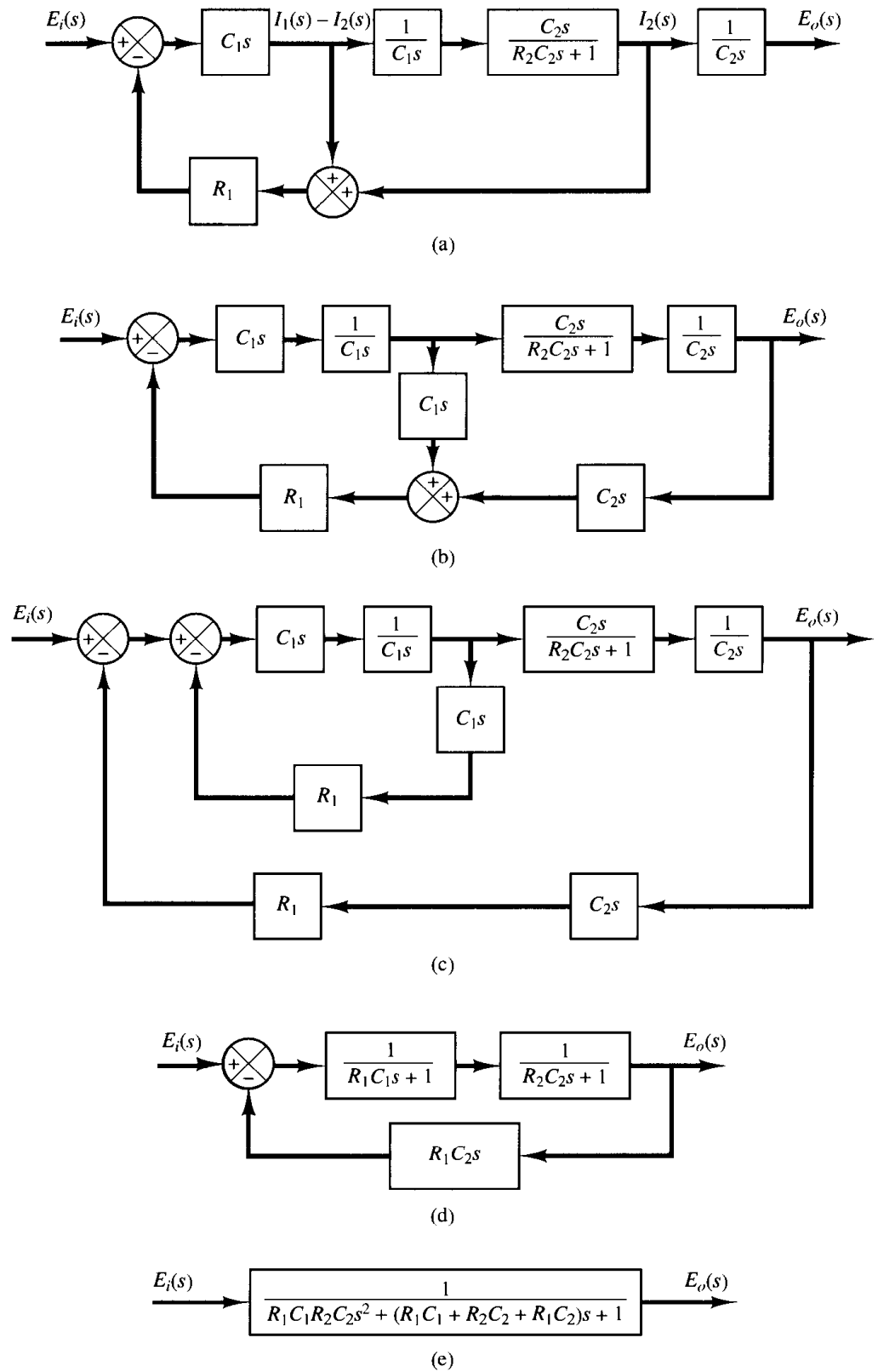
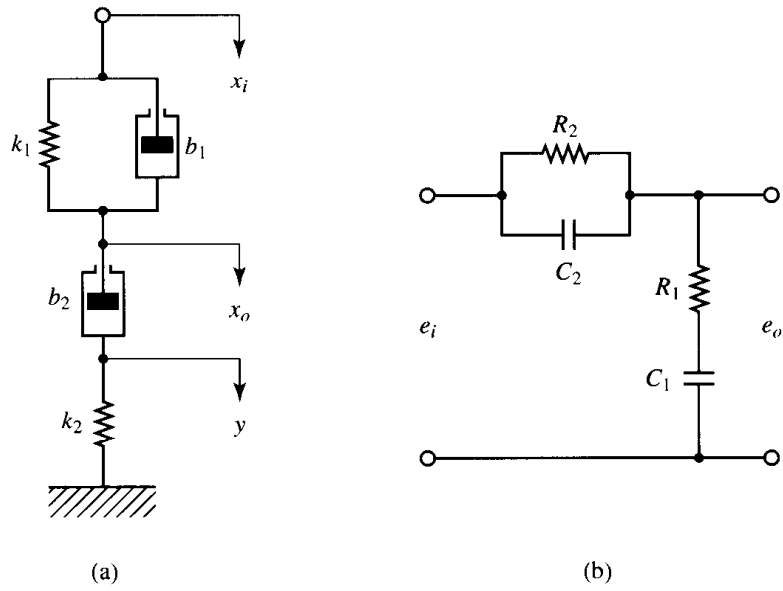


Figure 3-39

Block diagrams for the system shown in Figure 3-37. (a) through (e) show successive simplifications of block diagrams.

Figure 3–40

(a) Mechanical system; (b) analogous electrical system.



By taking the Laplace transforms of these two equations, assuming zero initial conditions, we have

$$b_1[sX_i(s) - sX_o(s)] + k_1[X_i(s) - X_o(s)] = b_2[sX_o(s) - sY(s)]$$

$$b_2[sX_o(s) - sY(s)] = k_2Y(s)$$

If we eliminate $Y(s)$ from the last two equations, then we obtain

$$b_1[sX_i(s) - sX_o(s)] + k_1[X_i(s) - X_o(s)] = b_2sX_o(s) - b_2s \frac{b_2sX_o(s)}{b_2s + k_2}$$

or

$$(b_1s + k_1)X_i(s) = \left(b_1s + k_1 + b_2s - b_2s \frac{b_2s}{b_2s + k_2} \right) X_o(s)$$

Hence the transfer function $X_o(s)/X_i(s)$ can be obtained as

$$\frac{X_o(s)}{X_i(s)} = \frac{\left(\frac{b_1}{k_1}s + 1 \right) \left(\frac{b_2}{k_2}s + 1 \right)}{\left(\frac{b_1}{k_1}s + 1 \right) \left(\frac{b_2}{k_2}s + 1 \right) + \frac{b_2}{k_1}s}$$

For the electrical system shown in Figure 3–40(b), the transfer function $E_o(s)/E_i(s)$ is found to be

$$\frac{E_o(s)}{E_i(s)} = \frac{R_1 + \frac{1}{C_1s}}{\frac{1}{(1/R_2) + C_2s} + R_1 + \frac{1}{C_1s}}$$

$$= \frac{(R_1C_1s + 1)(R_2C_2s + 1)}{(R_1C_1s + 1)(R_2C_2s + 1) + R_2C_1s}$$

A comparison of the transfer functions shows that the systems shown in Figures 3–40(a) and (b) are analogous.

- A-3-13.** In the liquid-level system of Figure 3-41, assume that the outflow rate Q m³/sec through the outflow valve is related to the head H m by

$$Q = K\sqrt{H} = 0.01 \sqrt{H}$$

Assume also that when the inflow rate Q_i is 0.015 m³/sec the head stays constant. At $t = 0$ the inflow valve is closed and so there is no inflow for $t \geq 0$. Find the time necessary to empty the tank to half the original head. The capacitance C of the tank is 2 m².

Solution. When the head is stationary, the inflow rate equals the outflow rate. Thus head H_o at $t = 0$ is obtained from

$$0.015 = 0.01 \sqrt{H_o}$$

or

$$H_o = 2.25 \text{ m}$$

The equation for the system for $t > 0$ is

$$-C dH = Q dt$$

or

$$\frac{dH}{dt} = -\frac{Q}{C} = \frac{-0.01 \sqrt{H}}{2}$$

Hence

$$\frac{dH}{\sqrt{H}} = -0.005 dt$$

Assume that, at $t = t_1$, $H = 1.125$ m. Integrating both sides of this last equation, we obtain

$$\int_{2.25}^{1.125} \frac{dH}{\sqrt{H}} = \int_0^{t_1} (-0.005) dt = -0.005t_1$$

It follows that

$$2\sqrt{H} \Big|_{2.25}^{1.125} = 2\sqrt{1.125} - 2\sqrt{2.25} = -0.005t_1$$

or

$$t_1 = 175.7$$

Thus, the head becomes half the original value (2.25 m) in 175.7 sec.

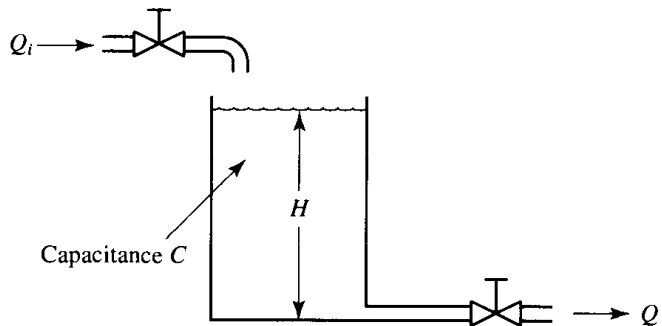


Figure 3-41
Liquid-level system.

A-3-14. Consider the liquid-level system shown in Figure 3-42. At steady state, the inflow rate and outflow rate are both $\bar{Q}$ and the flow rate between the tanks is zero. The heads of tanks 1 and 2 are both $\bar{H}$. At $t = 0$, the inflow rate is changed from $\bar{Q}$ to $\bar{Q} + q$, where q is a small change in the inflow rate. The resulting changes in the heads (h_1 and h_2) and flow rates (q_1 and q_2) are assumed to be small. The capacitances of tanks 1 and 2 are C_1 and C_2 , respectively. The resistance of the valve between the tanks is R_1 and that of the outflow valve is R_2 .

Derive mathematical models for the system when (a) q is the input and h_2 the output, (b) q is the input and q_2 the output, and (c) q is the input and h_1 the output.

Solution. (a) For tank 1, we have

$$C_1 dh_1 = q_1 dt$$

where

$$q_1 = \frac{h_2 - h_1}{R_1}$$

Consequently,

$$R_1 C_1 \frac{dh_1}{dt} + h_1 = h_2 \quad (3-98)$$

For tank 2, we get

$$C_2 dh_2 = (q - q_1 - q_2) dt$$

where

$$q_1 = \frac{h_2 - h_1}{R_1}, \quad q_2 = \frac{h_2}{R_2}$$

It follows that

$$R_2 C_2 \frac{dh_2}{dt} + \frac{R_2}{R_1} h_2 + h_2 = R_2 q + \frac{R_2}{R_1} h_1 \quad (3-99)$$

By eliminating h_1 from Equations (3-98) and (3-99), we have

$$R_1 C_1 R_2 C_2 \frac{d^2 h_2}{dt^2} + (R_1 C_1 + R_2 C_2 + R_2 C_1) \frac{dh_2}{dt} + h_2 = R_1 R_2 C_1 \frac{dq}{dt} + R_2 q \quad (3-100)$$

In terms of the transfer function, we have

$$\frac{H_2(s)}{Q(s)} = \frac{R_2(R_1 C_1 s + 1)}{R_1 C_1 R_2 C_2 s^2 + (R_1 C_1 + R_2 C_2 + R_2 C_1)s + 1}$$

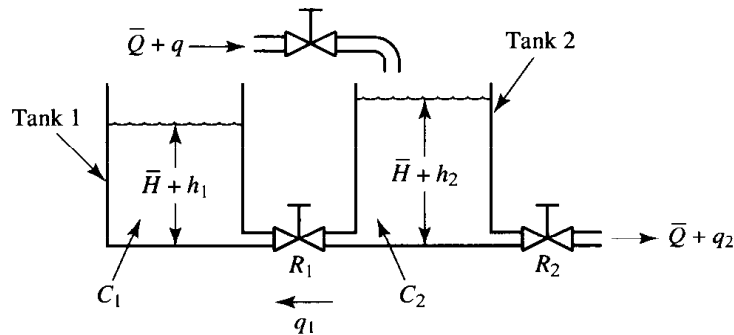


Figure 3-42
Liquid-level system.

This is the desired mathematical model in which q is considered the input and h_2 is the output. (b) Substitution of $h_2 = R_2 q_2$ into Equation (3-100) gives

$$R_1 C_1 R_2 C_2 \frac{d^2 q_2}{dt^2} + (R_1 C_1 + R_2 C_2 + R_2 C_1) \frac{dq_2}{dt} + q_2 = R_1 C_1 \frac{dq}{dt} + q$$

This equation is a mathematical model of the system when q is considered the input and q_2 is the output. In terms of the transfer function, we obtain

$$\frac{Q_2(s)}{Q(s)} = \frac{R_1 C_1 s + 1}{R_1 C_1 R_2 C_2 s^2 + (R_1 C_1 + R_2 C_2 + R_2 C_1) s + 1}$$

(c) Elimination of h_2 from Equations (3-98) and (3-99) yields

$$R_1 C_1 R_2 C_2 \frac{d^2 h_1}{dt^2} + (R_1 C_1 + R_2 C_2 + R_2 C_1) \frac{dh_1}{dt} + h_1 = R_2 q$$

which is a mathematical model of the system in which q is considered the input and h_1 is the output. In terms of the transfer function, we get

$$\frac{H_1(s)}{Q(s)} = \frac{R_2}{R_1 C_1 R_2 C_2 s^2 + (R_1 C_1 + R_2 C_2 + R_2 C_1) s + 1}$$

A-3-15. Consider the liquid-level system shown in Figure 3-43. In the system, $\bar{Q}_1$ and $\bar{Q}_2$ are steady-state inflow rates and $\bar{H}_1$ and $\bar{H}_2$ are steady-state heads. The quantities q_{i1} , q_{i2} , h_1 , h_2 , q_1 , and q_o are considered small. Obtain a state-space representation for the system when h_1 and h_2 are the outputs and q_{i1} and q_{i2} are the inputs.

Solution. The equations for the system are

$$C_1 dh_1 = (q_{i1} - q_1) dt \quad (3-101)$$

$$\frac{h_1 - h_2}{R_1} = q_1 \quad (3-102)$$

$$C_2 dh_2 = (q_1 + q_{i2} - q_o) dt \quad (3-103)$$

$$\frac{h_2}{R_2} = q_o \quad (3-104)$$

Elimination of q_1 from Equation (3-101) using Equation (3-102) results in

$$\frac{dh_1}{dt} = \frac{1}{C_1} \left(q_{i1} - \frac{h_1 - h_2}{R_1} \right) \quad (3-105)$$

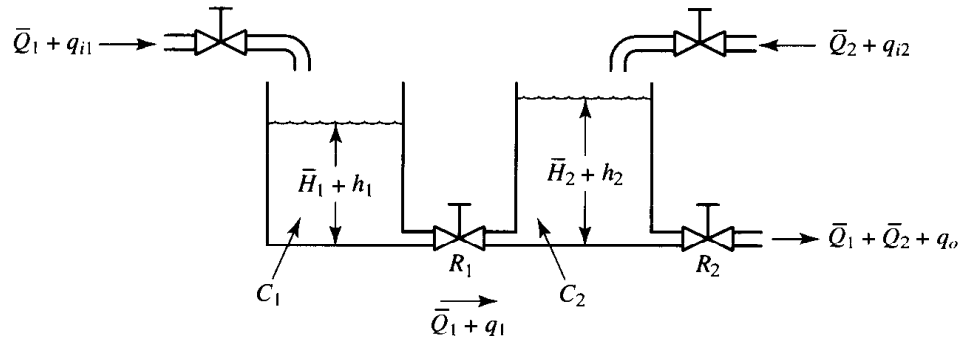


Figure 3-43
Liquid-level system.

Eliminating q_1 and q_o from Equation (3-103) by using Equations (3-102) and (3-104) gives

$$\frac{dh_2}{dt} = \frac{1}{C_2} \left(\frac{h_1 - h_2}{R_1} + q_{i2} - \frac{h_2}{R_2} \right) \quad (3-106)$$

Define state variables x_1 and x_2 by

$$x_1 = h_1$$

$$x_2 = h_2$$

the input variables u_1 and u_2 by

$$u_1 = q_{i1}$$

$$u_2 = q_{i2}$$

and the output variables y_1 and y_2 by

$$y_1 = h_1 = x_1$$

$$y_2 = h_2 = x_2$$

Then Equations (3-105) and (3-106) can be written as

$$\begin{aligned} \dot{x}_1 &= -\frac{1}{R_1 C_1} x_1 + \frac{1}{R_1 C_1} x_2 + \frac{1}{C_1} u_1 \\ \dot{x}_2 &= \frac{1}{R_1 C_2} x_1 - \left(\frac{1}{R_1 C_2} + \frac{1}{R_2 C_2} \right) x_2 + \frac{1}{C_2} u_2 \end{aligned}$$

In the form of the standard vector-matrix representation, we have

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -\frac{1}{R_1 C_1} & \frac{1}{R_1 C_1} \\ \frac{1}{R_1 C_2} & -\left(\frac{1}{R_1 C_2} + \frac{1}{R_2 C_2} \right) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} \frac{1}{C_1} & 0 \\ 0 & \frac{1}{C_2} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$$

which is the state equation, and

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

which is the output equation.

- A-3-16.** Considering small deviations from steady-state operation, draw a block diagram of the air heating system shown in Figure 3-44. Assume that the heat loss to the surroundings and the heat capacitance of the metal parts of the heater are negligible.

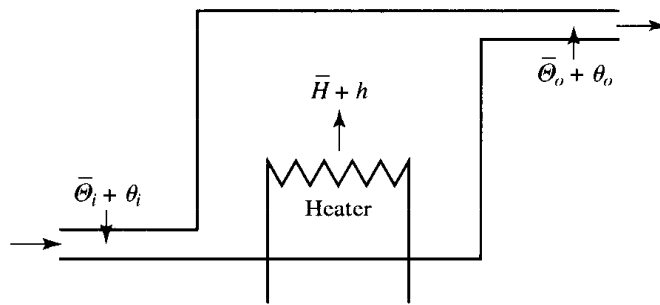


Figure 3-44
Air heating system.

Solution. Let us define

$\bar{\theta}_i$ = steady-state temperature of inlet air, °C

$\bar{\theta}_o$ = steady-state temperature of outlet air, °C

G = mass flow rate of air through the heating chamber, kg/sec

M = mass of air contained in the heating chamber, kg

c = specific heat of air, kcal/kg °C

R = thermal resistance, °C sec/kcal

C = thermal capacitance of air contained in the heating chamber = Mc , kcal/°C

$\bar{H}$ = steady-state heat input, kcal/sec

Let us assume that the heat input is suddenly changed from $\bar{H}$ to $\bar{H} + h$ and the inlet air temperature is suddenly changed from $\bar{\theta}_i$ to $\bar{\theta}_i + \theta_i$. Then the outlet air temperature will be changed from $\bar{\theta}_o$ to $\bar{\theta}_o + \theta_o$.

The equation describing the system behavior is

$$C d\theta_o = [h + Gc(\theta_i - \theta_o)] dt$$

or

$$C \frac{d\theta_o}{dt} = h + Gc(\theta_i - \theta_o)$$

Noting that

$$Gc = \frac{1}{R}$$

we obtain

$$C \frac{d\theta_o}{dt} = h + \frac{1}{R}(\theta_i - \theta_o)$$

or

$$RC \frac{d\theta_o}{dt} + \theta_o = Rh + \theta_i$$

Taking the Laplace transforms of both sides of this last equation and substituting the initial condition that $\theta_o(0) = 0$, we obtain

$$\Theta_o(s) = \frac{R}{RCs + 1} H(s) + \frac{1}{RCs + 1} \Theta_i(s)$$

The block diagram of the system corresponding to this equation is shown in Figure 3–45.

- A-3-17.** Consider the thin, glass-wall, mercury thermometer system shown in Figure 3–46. Assume that the thermometer is at a uniform temperature $\bar{\theta}$ °C (ambient temperature) and that at $t = 0$ it is immersed in a bath of temperature $\bar{\theta} + \theta_b$ °C, where θ_b is the bath temperature (which may be constant or changing) measured from the ambient temperature $\bar{\theta}$. Define the instantaneous thermometer temperature by $\bar{\theta} + \theta$ °C so that θ is the change in the thermometer temperature satisfying the condition that $\theta(0) = 0$. Obtain a mathematical model for the system. Also obtain an electrical analog of the thermometer system.

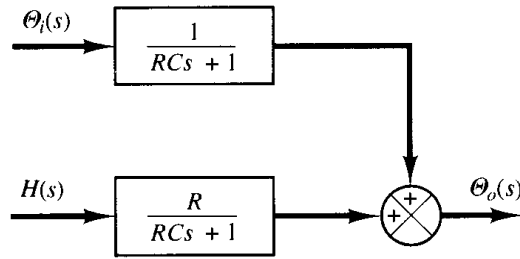


Figure 3-45
Block diagram of the air heating system shown in Figure 3-44.

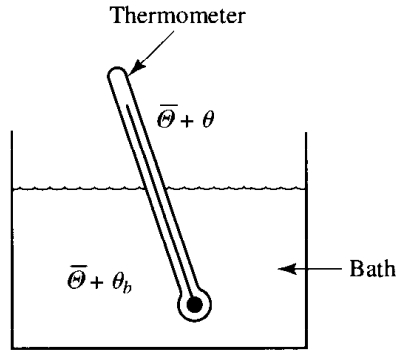


Figure 3-46
Thin, glass-wall, mercury thermometer system.

Solution. A mathematical model for the system can be derived by considering heat balance as follows: The heat entering the thermometer during dt sec is $q dt$, where q is the heat flow rate to the thermometer. This heat is stored in the thermal capacitance C of the thermometer, thereby raising its temperature by $d\theta$. Thus the heat-balance equation is

$$C d\theta = q dt \quad (3-107)$$

Since thermal resistance R may be written as

$$R = \frac{d(\Delta\theta)}{dq} = \frac{\Delta\theta}{q}$$

heat flow rate q may be given, in terms of thermal resistance R , as

$$q = \frac{(\bar{\Theta} + \theta_b) - (\bar{\Theta} + \theta)}{R} = \frac{\theta_b - \theta}{R}$$

where $\bar{\Theta} + \theta_b$ is the bath temperature and $\bar{\Theta} + \theta$ is the thermometer temperature. Hence, we can rewrite Equation (3-107) as

$$C \frac{d\theta}{dt} = \frac{\theta_b - \theta}{R}$$

or

$$RC \frac{d\theta}{dt} + \theta = \theta_b \quad (3-108)$$

Equation (3-108) is a mathematical model of the thermometer system.

Referring to Equation (3-108), an electrical analog for the thermometer system can be written as

$$RC \frac{de_o}{dt} + e_o = e_i$$

An electrical circuit represented by this last equation is shown in Figure 3-47.

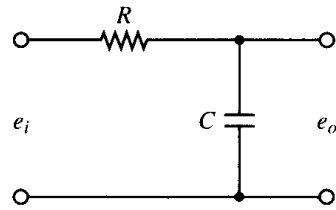


Figure 3-47
Electrical analog of the thermometer system shown in Figure 3-46.

A-3-18. Linearize the nonlinear equation

$$z = xy$$

in the region $5 \leq x \leq 7, 10 \leq y \leq 12$. Find the error if the linearized equation is used to calculate the value of z when $x = 5, y = 10$.

Solution. Since the region considered is given by $5 \leq x \leq 7, 10 \leq y \leq 12$, choose $\bar{x} = 6, \bar{y} = 11$. Then $\bar{z} = \bar{x}\bar{y} = 66$. Let us obtain a linearized equation for the nonlinear equation near a point $\bar{x} = 6, \bar{y} = 11$.

Expanding the nonlinear equation into a Taylor series about point $x = \bar{x}, y = \bar{y}$ and neglecting the higher-order terms, we have

$$z - \bar{z} = a(x - \bar{x}) + b(y - \bar{y})$$

where

$$a = \left. \frac{\partial(xy)}{\partial x} \right|_{x=\bar{x}, y=\bar{y}} = \bar{y} = 11$$

$$b = \left. \frac{\partial(xy)}{\partial y} \right|_{x=\bar{x}, y=\bar{y}} = \bar{x} = 6$$

Hence the linearized equation is

$$z - 66 = 11(x - 6) + 6(y - 11)$$

or

$$z = 11x + 6y - 66$$

When $x = 5, y = 10$, the value of z given by the linearized equation is

$$z = 11x + 6y - 66 = 55 + 60 - 66 = 49$$

The exact value of z is $z = xy = 50$. The error is thus $50 - 49 = 1$. In terms of percentage, the error is 2%.

A-3-19. Consider the liquid-level system shown in Figure 3-48. At steady state the inflow rate is $Q_i = \bar{Q}$, the outflow rate is $Q_o = \bar{Q}$, and head is $H = \bar{H}$. If the flow is turbulent, then we have

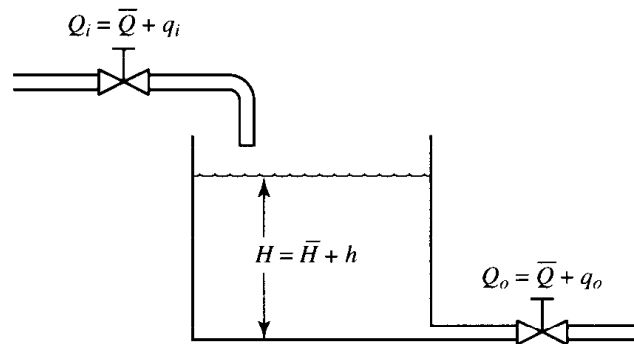


Figure 3-48
Liquid-level system.

$$\bar{Q} = K\sqrt{\bar{H}}$$

Assume that at $t = 0$ the inflow rate is changed from $Q_i = \bar{Q}$ to $Q_i = \bar{Q} + q_i$. This change causes the head to change from $H = \bar{H}$ to $H = \bar{H} + h$, which, in turn, causes the outflow rate to change from $Q_o = \bar{Q}$ to $Q_o = \bar{Q} + q_o$. For this system we have

$$C \frac{dH}{dt} = Q_i - Q_o = Q_i - K\sqrt{H}$$

where C is the capacitance of the tank. Let us define

$$\frac{dH}{dt} = f(H, Q_i) = \frac{1}{C} Q_i - \frac{K\sqrt{H}}{C} \quad (3-109)$$

Note that the steady-state operating condition is $(\bar{H}, \bar{Q})$ and $H = \bar{H} + h$, $Q_i = \bar{Q} + q_i$. Since at steady-state operation $dH/dt = 0$, we have $f(\bar{H}, \bar{Q}) = 0$.

Linearize Equation (3-109) near the operating point $(\bar{H}, \bar{Q})$.

Solution. Using the linearization technique presented in Section 3-10, a linearized equation for Equation (3-109) can be obtained as follows:

$$\frac{dH}{dt} - f(\bar{H}, \bar{Q}) = \frac{\partial f}{\partial H} (H - \bar{H}) + \frac{\partial f}{\partial Q_i} (Q_i - \bar{Q}) \quad (3-110)$$

where

$$f(\bar{H}, \bar{Q}) = 0$$

$$\left. \frac{\partial f}{\partial H} \right|_{H=\bar{H}, Q_i=\bar{Q}} = -\frac{K}{2C\sqrt{\bar{H}}} = -\frac{\bar{Q}}{\sqrt{\bar{H}}} \frac{1}{2C\sqrt{\bar{H}}} = -\frac{\bar{Q}}{2C\bar{H}} = -\frac{1}{RC}$$

where we used the resistance R defined by

$$R = \frac{2\bar{H}}{\bar{Q}}$$

Also,

$$\left. \frac{\partial f}{\partial Q_i} \right|_{H=\bar{H}, Q_i=\bar{Q}} = \frac{1}{C}$$

Then Equation (3-110) can be written as

$$\frac{dH}{dt} = -\frac{1}{RC} (H - \bar{H}) + \frac{1}{C} (Q_i - \bar{Q}) \quad (3-111)$$

Since $H - \bar{H} = h$ and $Q_i - \bar{Q} = q_i$, Equation (3-111) can be written as

$$\frac{dh}{dt} = -\frac{1}{RC} h + \frac{1}{C} q_i$$

or

$$RC \frac{dh}{dt} + h = Rq_i$$

which is the linearized equation for the liquid-level system and is the same as Equation (3-69) that we obtained in Section 3-8.

A-3-20. Consider the hydraulic servo system shown in Figure 3-49. Assuming that the load reaction forces are not negligible, derive a mathematical model of the system. Assume also that the mass of the power piston is included in the load mass m .

Solution. In deriving a mathematical model of the system when the load reactive forces are not negligible, such effects as the pressure drop across the orifice, the leakage of oil around the valve and around the piston, and the compressibility of the oil must be considered.

The pressure drop across the orifice is a function of the supply pressure p_s and the pressure difference $\Delta p = p_1 - p_2$. Thus the flow rate q is a nonlinear function of valve displacement x and pressure difference Δp or

$$q = f(x, \Delta p)$$

Linearizing this nonlinear equation about the origin ($x = 0, \Delta p = 0, q = 0$), we obtain, referring to Equation (3-82),

$$q = K_1 x - K_2 \Delta p \quad (3-112)$$

The flow rate q can be considered as consisting of three parts

$$q = q_0 + q_L + q_C \quad (3-113)$$

where q_0 = useful flow rate to the power cylinder causing power piston to move, kg/sec

q_L = leakage flow rate, kg/sec

q_C = equivalent compressibility flow rate, kg/sec

Let us obtain specific expressions for q_0 , q_L , and q_C . The flow $q_0 dt$ to the left-hand side of the power piston causes the piston to move to the right by dy . So we have

$$A \rho dy = q_0 dt$$

where A (m^2) is the power piston area, ρ (kg/m^3) the density of oil, and dy (m) the displacement of the power piston. Then

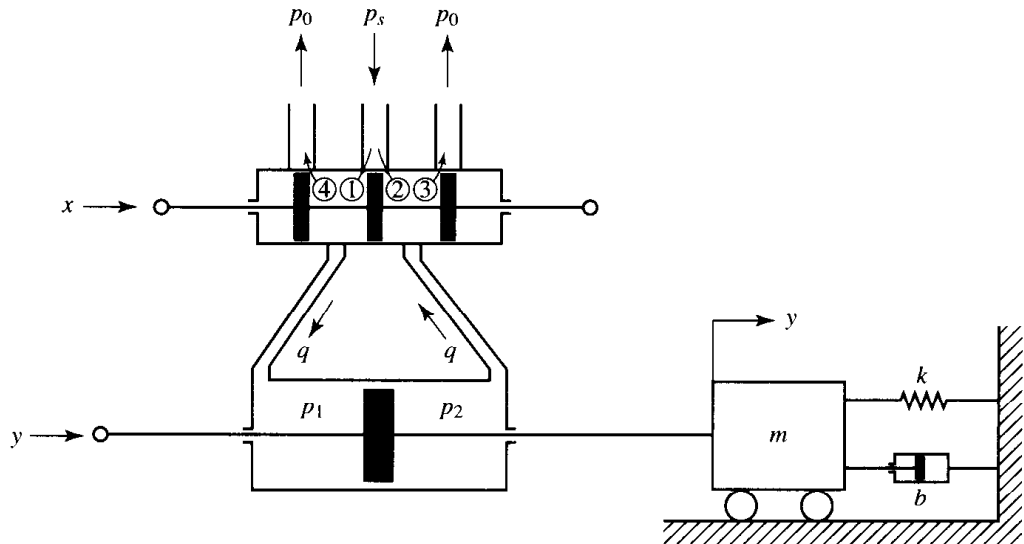


Figure 3-49
Hydraulic servo
system.

$$q_0 = A\varrho \frac{dy}{dt} \quad (3-114)$$

The leakage component q_L can be written

$$q_L = L \Delta p \quad (3-115)$$

where L is the leakage coefficient of the system.

The equivalent compressibility flow rate q_C can be expressed in terms of the effective bulk modulus K of oil (including the effects of entrapped air, expansion of pipes, etc.), where

$$K = \frac{d \Delta p}{-dV/V}$$

(Here dV is negative and so $-dV$ is positive.) Rewriting this last equation gives

$$-dV = \frac{V}{K} d \Delta p$$

or

$$\varrho \frac{-dV}{dt} = \frac{\varrho V}{K} \frac{d \Delta p}{dt}$$

Noting that $q_C = \varrho(-dV)/dt$, we find

$$q_C = \frac{\varrho V}{K} \frac{d \Delta p}{dt} \quad (3-116)$$

where V is the effective volume of oil under compression (that is, approximately half the total power cylinder volume).

Using Equations (3-112) through (3-116),

$$q = K_1 x - K_2 \Delta p = A\varrho \frac{dy}{dt} + L \Delta p + \frac{\varrho V}{K} \frac{d \Delta p}{dt}$$

or

$$A\varrho \frac{dy}{dt} + \frac{\varrho V}{K} \frac{d \Delta p}{dt} + (L + K_2) \Delta p = K_1 x \quad (3-117)$$

The force developed by the power piston is $A \Delta p$, and this force is applied to the load elements. Thus

$$m \frac{d^2 y}{dt^2} + b \frac{dy}{dt} + ky = A \Delta p \quad (3-118)$$

Eliminating Δp from Equations (3-117) and (3-118) results in

$$\begin{aligned} \frac{\varrho V m}{KA} \frac{d^3 y}{dt^3} + \left[\frac{\varrho V b}{KA} + \frac{(L + K_2)m}{A} \right] \frac{d^2 y}{dt^2} \\ + \left[A\varrho + \frac{\varrho V k}{KA} + \frac{(L + K_2)b}{A} \right] \frac{dy}{dt} + \frac{(L + K_2)k}{A} y = K_1 x \end{aligned}$$

This is a mathematical model of the system relating the valve spool displacement x and the power piston displacement y when the load reactive forces are not negligible.

PROBLEMS

B-3-1. Simplify the block diagram shown in Figure 3-50 and obtain the closed-loop transfer function $C(s)/R(s)$.

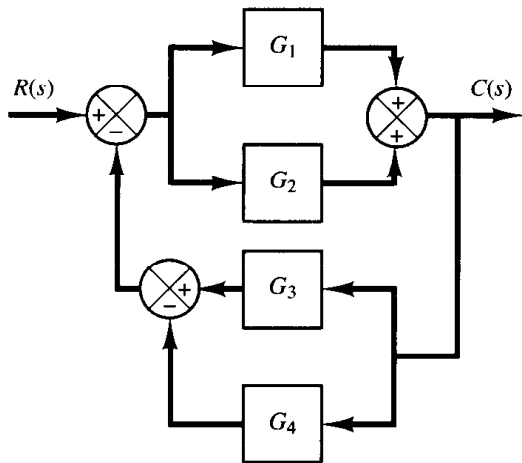


Figure 3-50 Block diagram of a system.

B-3-2. Simplify the block diagram shown in Figure 3-51 and obtain the transfer function $C(s)/R(s)$.

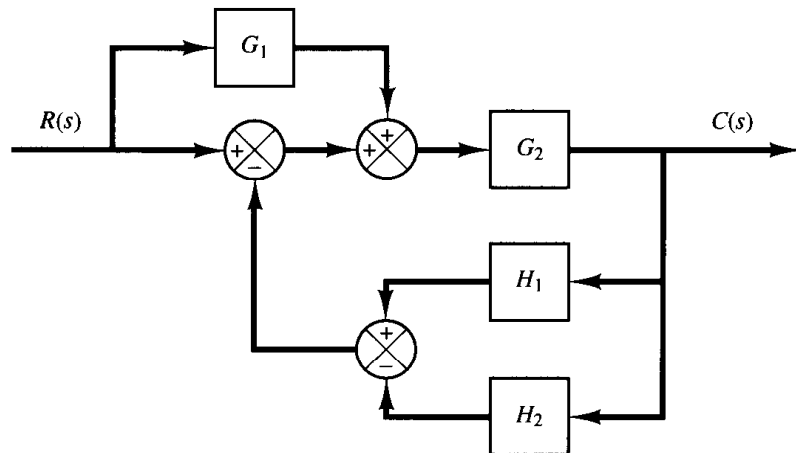


Figure 3-51 Block diagram of a system.

B-3-3. Simplify the block diagram shown in Figure 3-52 and obtain the closed-loop transfer function $C(s)/R(s)$.

B-3-4. Obtain a state-space representation of the system shown in Figure 3-53.

B-3-5. Consider the system described by

$$\ddot{y} + 3\dot{y} + 2y = u$$

Derive a state-space representation of the system.

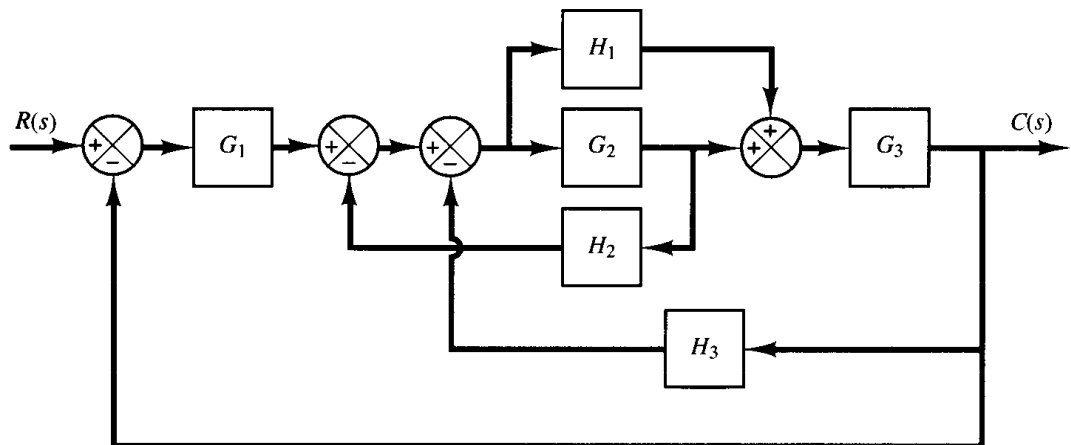


Figure 3-52 Block diagram of a system.

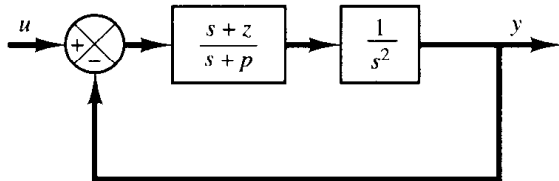


Figure 3-53 Control system.

B-3-6. Consider the system described by

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -4 & -1 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Obtain the transfer function of the system.

B-3-7. Obtain the transfer function $X_o(s)/X_i(s)$ of each of the three mechanical systems shown in Figure 3-54. In the diagrams, x_i denotes the input displacement and x_o denotes the output displacement. (Each displacement is measured from its equilibrium position.)

B-3-8. Obtain mathematical models of the mechanical systems shown in Figures 3-55(a) and (b).

B-3-9. Obtain a state-space representation of the mechanical system shown in Figure 3-56, where u_1 and u_2 are the inputs and y_1 and y_2 are the outputs.

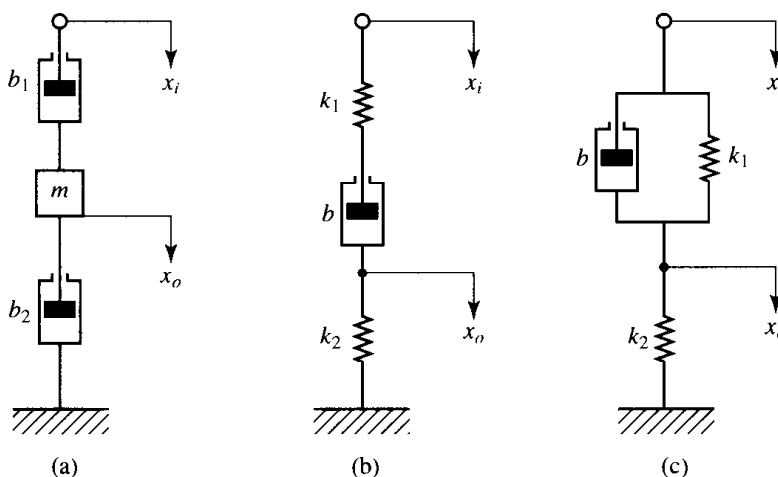


Figure 3-54 Mechanical systems.

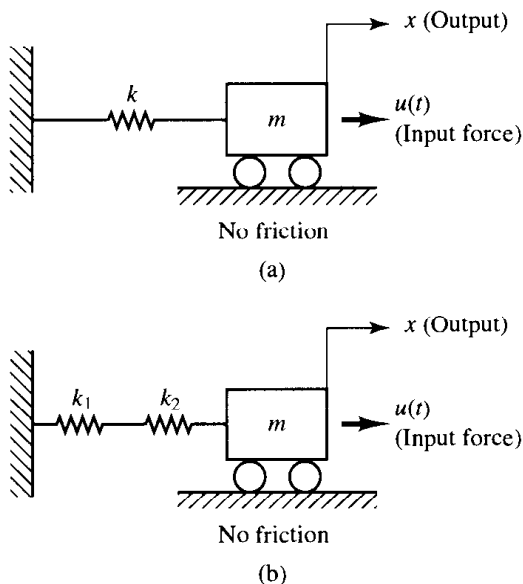


Figure 3-55 Mechanical systems.

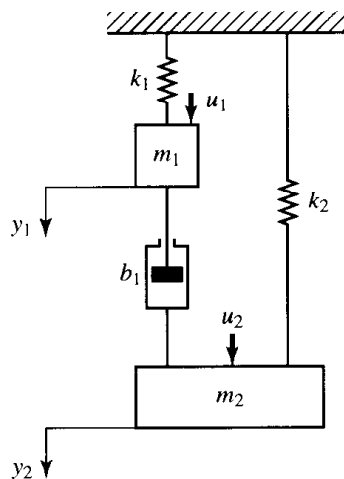


Figure 3-56
Mechanical system.

B-3-10. Consider the spring-loaded pendulum system shown in Figure 3-57. Assume that the spring force acting on the pendulum is zero when the pendulum is vertical, or $\theta = 0$. Assume also that the friction involved is negligible and the angle of oscillation θ is small. Obtain a mathematical model of the system.

B-3-11. Referring to Example 3-4, consider the inverted pendulum system shown in Figure 3-58. Assume that the mass of the inverted pendulum is m and is evenly distributed along the length of the rod. (The center of gravity of the pendulum is located at the center of the rod.) Assuming that θ is small, derive mathematical models for the system in the forms of differential equations, transfer functions, and state-space equations.

B-3-12. Derive the transfer function of the electrical system shown in Figure 3-59. Draw a schematic diagram of an analogous mechanical system.

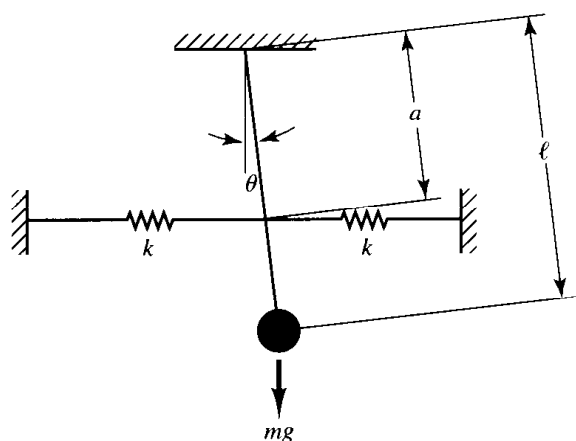


Figure 3-57 Spring-loaded pendulum system.

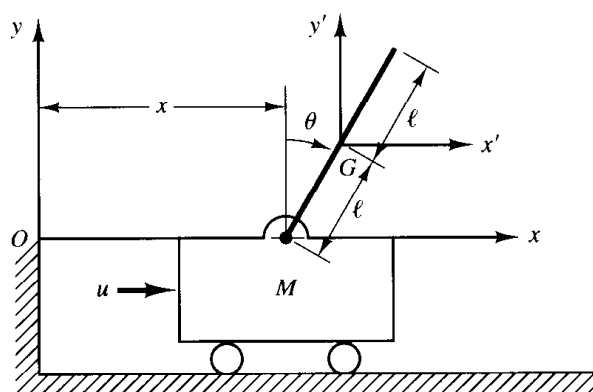


Figure 3-58 Inverted pendulum system.

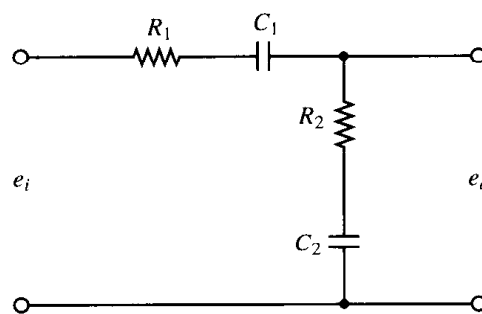


Figure 3-59 Electrical system.

B-3-13. Consider the liquid-level system shown in Figure 3-60. Assuming that $\bar{H} = 3$ m, $\bar{Q} = 0.02$ m³/sec, and the cross-sectional area of the tank is equal to 5 m², obtain the time constant of the system at the operating point $(\bar{H}, \bar{Q})$. Assume that the flow through the valve is turbulent.

B-3-14. Consider the conical water tank system shown in Figure 3-61. The flow through the valve is turbulent and is related to the head H by

$$Q = 0.005 \sqrt{H}$$

where Q is the flow rate measured in m³/sec and H is in meters.

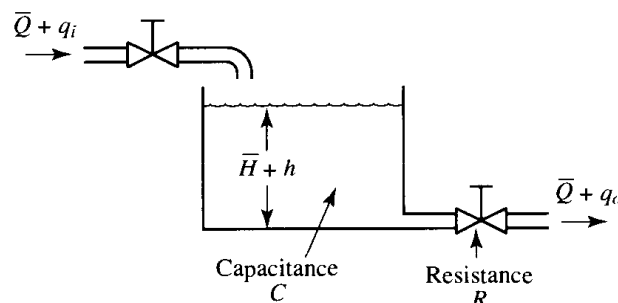


Figure 3-60 Liquid-level system.

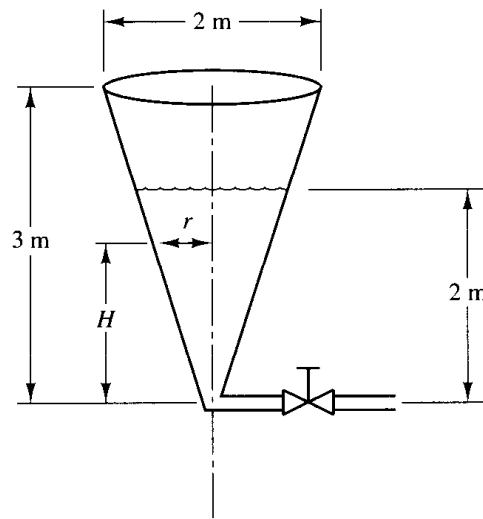


Figure 3-61 Conical water tank system.

Suppose that the head is 2 m at $t = 0$. What will be the head at $t = 60$ sec?

B-3-15. Consider the liquid-level system shown in Figure 3-62. At steady state the inflow rate is $\bar{Q}$ and the outflow rate is also $\bar{Q}$. Assume that at $t = 0$ the inflow rate is changed from $\bar{Q}$ to $\bar{Q} + q_i$, where q_i is a small quantity. The disturbance input is q_d , which is also a small quantity. Draw a block diagram of the system and simplify it to obtain $H_2(s)$ as a function of $Q_i(s)$ and $Q_d(s)$, where $H_2(s) = \mathcal{L}[h_2(t)]$, $Q_i(s) = \mathcal{L}[q_i(t)]$, and $Q_d(s) = \mathcal{L}[q_d(t)]$. The capacitances of tanks 1 and 2 are C_1 and C_2 , respectively.

B-3-16. A thermocouple has a time constant of 2 sec. A thermal well has a time constant of 30 sec. When the thermocouple is inserted into the well, this temperature-measuring device can be considered a two-capacitance system.

Determine the time constants of the combined thermocouple–thermal well system. Assume that the weight of the thermocouple is 8 g and the weight of the thermal well is 40 g. Assume also that the specific heats of the thermocouple and thermal well are the same.

B-3-17. Suppose that the flow rate Q and head H in a liquid-level system are related by

$$Q = 0.002 \sqrt{H}$$

Obtain a linearized mathematical model relating the flow rate and head near the steady-state operating point $(\bar{H}, \bar{Q})$, where $\bar{H} = 2.25$ m and $\bar{Q} = 0.003$ m³/sec.

B-3-18. Find a linearized equation for

$$y = 0.2x^3$$

about a point $\bar{x} = 2$.

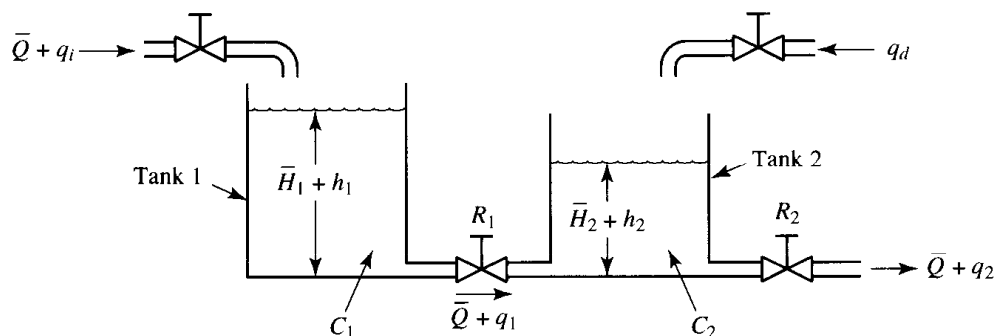


Figure 3-62 Liquid-level system.

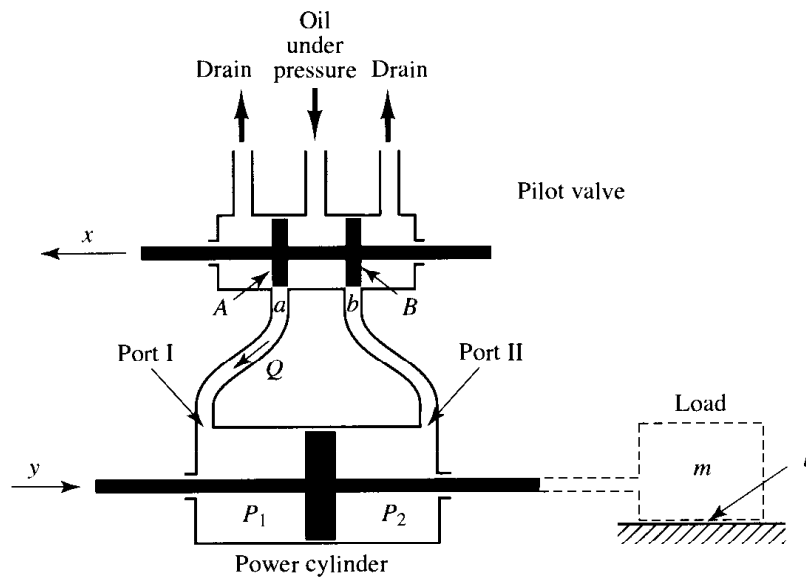


Figure 3–63 Schematic diagram of a hydraulic servomotor.

B-3-19. Linearize the nonlinear equation

$$z = x^2 + 4xy + 6y^2$$

in the region defined by $8 \leq x \leq 10$, $2 \leq y \leq 4$.

B-3-20. Consider the hydraulic servomotor shown in Figure 3–63. Derive the transfer function $Y(s)/X(s)$. Assume that the inertia force due to the mass of power piston and shaft is negligible compared with the inertia force due to the load mass m and viscous friction force $b\dot{y}$.

4

Transient-Response Analysis

4-1 INTRODUCTION

It was stated in Chapter 3 that the first step in analyzing a control system was to derive a mathematical model of the system. Once such a model is obtained, various methods are available for the analysis of system performance.

In practice, the input signal to a control system is not known ahead of time but is random in nature, and the instantaneous input cannot be expressed analytically. Only in some special cases is the input signal known in advance and expressible analytically or by curves, such as in the case of the automatic control of cutting tools.

In analyzing and designing control systems, we must have a basis of comparison of performance of various control systems. This basis may be set up by specifying particular test input signals and by comparing the responses of various systems to these input signals.

Many design criteria are based on such signals or on the response of systems to changes in initial conditions (without any test signals). The use of test signals can be justified because of a correlation existing between the response characteristics of a system to a typical test input signal and the capability of the system to cope with actual input signals.

Typical test signals. The commonly used test input signals are those of step functions, ramp functions, acceleration functions, impulse functions, sinusoidal functions, and the like. With these test signals, mathematical and experimental analyses of control systems can be carried out easily since the signals are very simple functions of time.

Which of these typical input signals to use for analyzing system characteristics may be determined by the form of the input that the system will be subjected to most frequently under normal operation. If the inputs to a control system are gradually changing functions of time, then a ramp function of time may be a good test signal. Similarly, if a system is subjected to sudden disturbances, a step function of time may be a good test signal; and for a system subjected to shock inputs, an impulse function may be best. Once a control system is designed on the basis of test signals, the performance of the system in response to actual inputs is generally satisfactory. The use of such test signals enables one to compare the performance of all systems on the same basis.

Transient response and steady-state response. The time response of a control system consists of two parts: the transient and the steady-state response. By transient response, we mean that which goes from the initial state to the final state. By steady-state response, we mean the manner in which the system output behaves as t approaches infinity.

Absolute stability, relative stability, and steady-state error. In designing a control system, we must be able to predict the dynamic behavior of the system from a knowledge of the components. The most important characteristic of the dynamic behavior of a control system is absolute stability, that is, whether the system is stable or unstable. A control system is in equilibrium if, in the absence of any disturbance or input, the output stays in the same state. A linear time-invariant control system is stable if the output eventually comes back to its equilibrium state when the system is subjected to an initial condition. A linear time-invariant control system is critically stable if oscillations of the output continue forever. It is unstable if the output diverges without bound from its equilibrium state when the system is subjected to an initial condition. Actually, the output of a physical system may increase to a certain extent but may be limited by mechanical “stops,” or the system may break down or become nonlinear after the output exceeds a certain magnitude so that the linear differential equations no longer apply.

Important system behavior (other than absolute stability) to which we must give careful consideration includes relative stability and steady-state error. Since a physical control system involves energy storage, the output of the system, when subjected to an input, cannot follow the input immediately but exhibits a transient response before a steady state can be reached. The transient response of a practical control system often exhibits damped oscillations before reaching a steady state. If the output of a system at steady state does not exactly agree with the input, the system is said to have steady-state error. This error is indicative of the accuracy of the system. In analyzing a control system, we must examine transient-response behavior and steady-state behavior.

Outline of the chapter. This chapter is concerned with system responses to aperiodic signals (such as step, ramp, acceleration, and impulse functions of time). The outline of the chapter is as follows: Section 4–1 has presented introductory material for the chapter. Section 4–2 treats the response of first-order systems to aperiodic inputs. Section 4–3 deals with the transient response of the second-order systems. Detailed analyses of the step response, ramp response, and impulse response of the second-order systems are presented. (The transient response analysis of higher-order systems is discussed in Chapter 5.) Section 4–4 gives an introduction to the MATLAB approach

to the solution of transient response. Section 4–5 presents an example of a transient-response problem solved with MATLAB.

4-2 FIRST-ORDER SYSTEMS

Consider the first-order system shown in Figure 4–1(a). Physically, this system may represent an RC circuit, thermal system, or the like. A simplified block diagram is shown in Figure 4–1(b). The input–output relationship is given by

$$\frac{C(s)}{R(s)} = \frac{1}{Ts + 1} \quad (4-1)$$

In the following, we shall analyze the system responses to such inputs as the unit-step, unit-ramp, and unit-impulse functions. The initial conditions are assumed to be zero.

Note that all systems having the same transfer function will exhibit the same output in response to the same input. For any given physical system, the mathematical response can be given a physical interpretation.

Unit-step response of first-order systems. Since the Laplace transform of the unit-step function is $1/s$, substituting $R(s) = 1/s$ into Equation (4–1), we obtain

$$C(s) = \frac{1}{Ts + 1} \frac{1}{s}$$

Expanding $C(s)$ into partial fractions gives

$$C(s) = \frac{1}{s} - \frac{T}{Ts + 1} = \frac{1}{s} - \frac{1}{s + (1/T)} \quad (4-2)$$

Taking the inverse Laplace transform of Equation (4–2), we obtain

$$c(t) = 1 - e^{-t/T}, \quad \text{for } t \geq 0 \quad (4-3)$$

Equation (4–3) states that initially the output $c(t)$ is zero and finally it becomes unity. One important characteristic of such an exponential response curve $c(t)$ is that at $t = T$ the value of $c(t)$ is 0.632, or the response $c(t)$ has reached 63.2% of its total change. This may be easily seen by substituting $t = T$ in $c(t)$. That is,

$$c(T) = 1 - e^{-1} = 0.632$$

Note that the smaller the time constant T , the faster the system response. Another important characteristic of the exponential response curve is that the slope of the tangent line at $t = 0$ is $1/T$, since

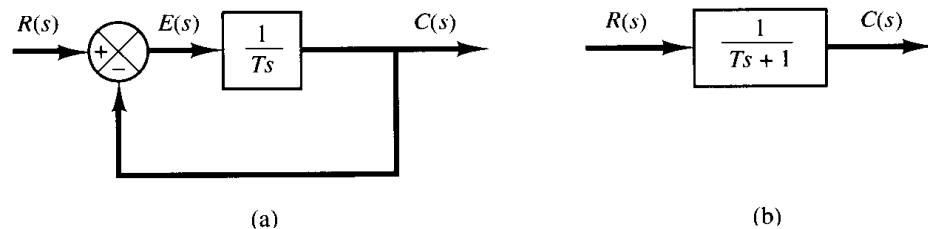


Figure 4–1
(a) Block diagram of a first-order system;
(b) simplified block diagram.

$$\left. \frac{dc}{dt} = \frac{1}{T} e^{-t/T} \right|_{t=0} = \frac{1}{T} \quad (4-4)$$

The output would reach the final value at $t = T$ if it maintained its initial speed of response. From Equation (4-4) we see that the slope of the response curve $c(t)$ decreases monotonically from $1/T$ at $t = 0$ to zero at $t = \infty$.

The exponential response curve $c(t)$ given by Equation (4-3) is shown in Figure 4-2. In one time constant, the exponential response curve has gone from 0 to 63.2% of the final value. In two time constants, the response reaches 86.5% of the final value. At $t = 3T$, $4T$, and $5T$, the response reaches 95%, 98.2%, and 99.3%, respectively, of the final value. Thus, for $t \geq 4T$, the response remains within 2% of the final value. As seen from Equation (4-3), the steady state is reached mathematically only after an infinite time. In practice, however, a reasonable estimate of the response time is the length of time the response curve needs to reach the 2% line of the final value, or four time constants.

Consider the system shown in Figure 4-3. To determine experimentally whether or not the system is of first order, plot the curve $\log |c(t) - c(\infty)|$, where $c(t)$ is the system output, as a function of t . If the curve turns out to be a straight line, the system is of first order. The time constant T can be read from the graph as the time T that satisfies the following equation:

$$c(T) - c(\infty) = 0.368 [c(0) - c(\infty)]$$

Note that instead of plotting $\log |c(t) - c(\infty)|$ versus t it is convenient to plot $|c(t) - c(\infty)|/|c(0) - c(\infty)|$ versus t on semilog paper, as shown in Figure 4-4.

Unit-ramp response of first-order systems. Since the Laplace transform of the unit-ramp function is $1/s^2$, we obtain the output of the system of Figure 4-1(a) as

$$C(s) = \frac{1}{Ts + 1} \frac{1}{s^2}$$

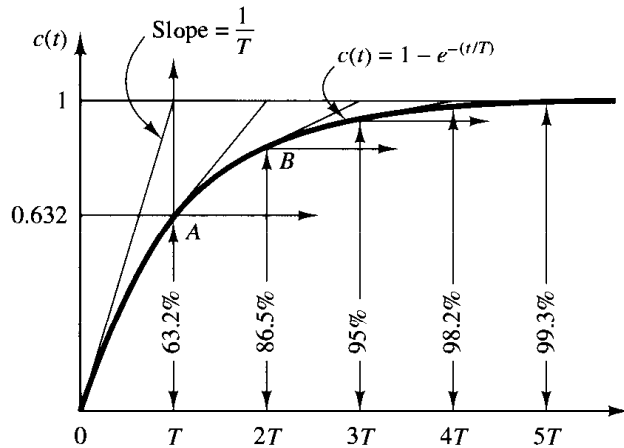


Figure 4-2
Exponential response curve.

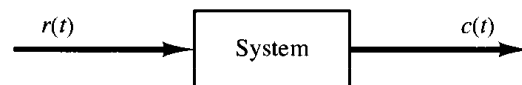


Figure 4-3
A general system.

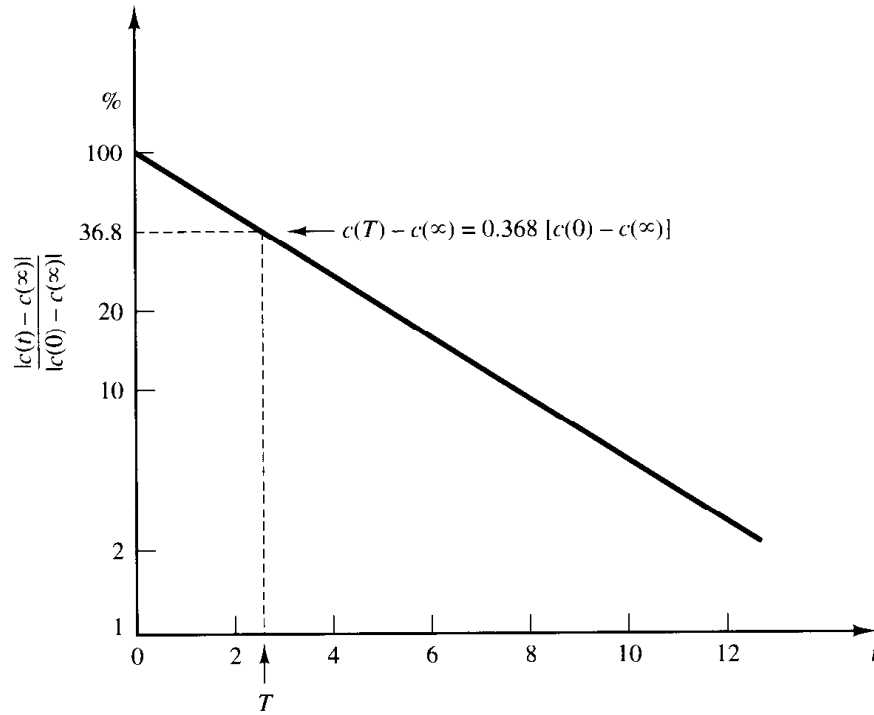


Figure 4-4
Plot of $|c(t) - c(\infty)| / |c(0) - c(\infty)|$ versus t on semilog paper.

Expanding $C(s)$ into partial fractions gives

$$C(s) = \frac{1}{s^2} - \frac{T}{s} + \frac{T^2}{Ts + 1} \quad (4-5)$$

Taking the inverse Laplace transform of Equation (4-5), we obtain

$$c(t) = t - T + Te^{-t/T}, \quad \text{for } t \geq 0$$

The error signal $e(t)$ is then

$$\begin{aligned} e(t) &= r(t) - c(t) \\ &= T(1 - e^{-t/T}) \end{aligned}$$

As t approaches infinity, $e^{-t/T}$ approaches zero, and thus the error signal $e(t)$ approaches T or

$$e(\infty) = T$$

The unit-ramp input and the system output are shown in Figure 4-5. The error in following the unit-ramp input is equal to T for sufficiently large t . The smaller the time constant T , the smaller the steady-state error in following the ramp input.

Unit-impulse response of first-order systems. For the unit-impulse input, $R(s) = 1$ and the output of the system of Figure 4-1(a) can be obtained as

$$C(s) = \frac{1}{Ts + 1}$$

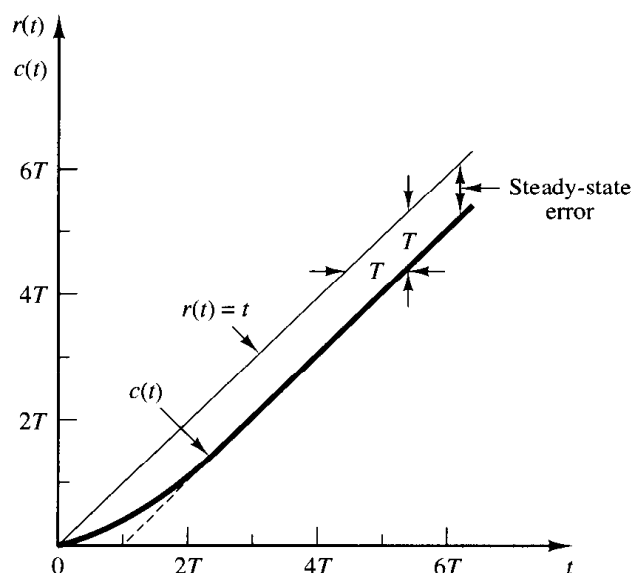


Figure 4-5
Unit-ramp response of the system
shown in Figure 4-1(a).

or

$$c(t) = \frac{1}{T} e^{-t/T}, \quad \text{for } t \geq 0 \quad (4-6)$$

The response curve given by Equation (4-6) is shown in Figure 4-6.

An important property of linear time-invariant systems. In the analysis above, it has been shown that for the unit-ramp input the output $c(t)$ is

$$c(t) = t - T + Te^{-t/T}, \quad \text{for } t \geq 0$$

For the unit-step input, which is the derivative of unit-ramp input, the output $c(t)$ is

$$c(t) = 1 - e^{-t/T}, \quad \text{for } t \geq 0$$

Finally, for the unit-impulse input, which is the derivative of unit-step input, the output $c(t)$ is

$$c(t) = \frac{1}{T} e^{-t/T}, \quad \text{for } t \geq 0$$

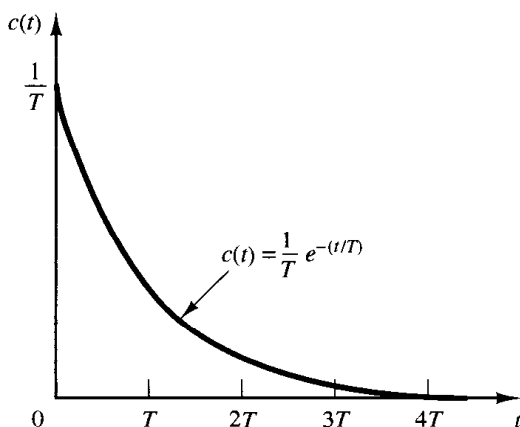


Figure 4-6
Unit-impulse response of the system
shown in Figure 4-1(a).

Comparison of the system responses to these three inputs clearly indicates that the response to the derivative of an input signal can be obtained by differentiating the response of the system to the original signal. It can also be seen that the response to the integral of the original signal can be obtained by integrating the response of the system to the original signal and by determining the integration constants from the zero output initial condition. This is a property of linear time-invariant systems. Linear time-varying systems and nonlinear systems do not possess this property.

EXAMPLE 4-1

Consider the liquid-level control system shown in Figure 4-7(a). (The controller is assumed to be a proportional controller; that is, the output of the controller is proportional to the input of the controller.) We assume that all the variables, r , q_i , h , and q_o are measured from their respective steady-state values $\bar{R}$, $\bar{Q}$, $\bar{H}$, and $\bar{Q}$. We also assume that the magnitudes of the variables r , q_i , h , and q_o are sufficiently small so that the system can be approximated by a linear mathematical model.

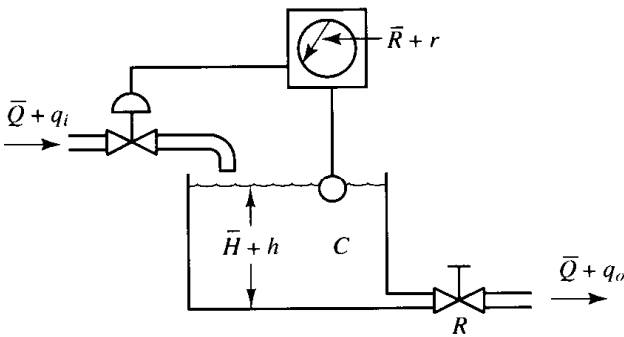
Referring to Section 3-8, we can obtain the transfer function of the liquid-level system as

$$\frac{H(s)}{Q_i(s)} = \frac{R}{RCs + 1}$$

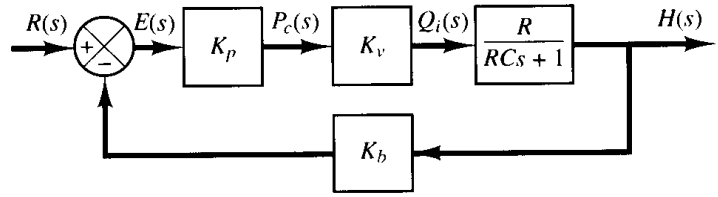
Since the controller is a proportional controller, the change in inflow q_i is proportional to the actuating error e so that $q_i = K_p K_v e$, where K_p is the gain of the controller and K_v is the gain of the control valve. In terms of Laplace-transformed quantities,

$$Q_i(s) = K_p K_v E(s)$$

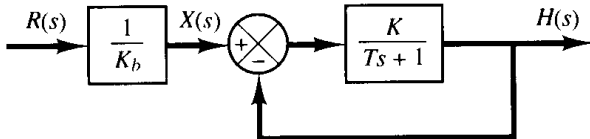
A block diagram of this system is shown in Figure 4-7(b). A simplified block diagram is given in Figure 4-7(c), where $X(s) = (1/K_b)R(s)$, $K = K_p K_v R K_b$, and $T = RC$.



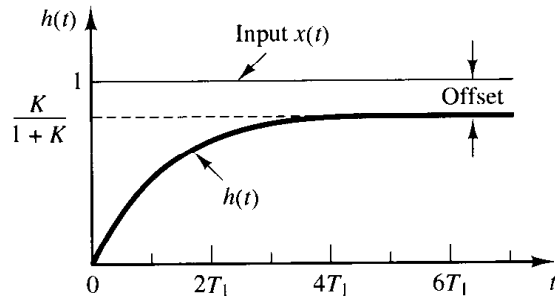
(a)



(b)



(c)



(d)

Figure 4-7

(a) Liquid-level control system; (b) block diagram; (c) simplified block diagram; (d) curve $h(t)$ versus t .

In what follows we shall investigate the response $h(t)$ to a change in the reference input. We shall assume a unit-step change in $x(t)$, where $x(t) = (1/K_b)r(t)$. The closed-loop transfer function between $H(s)$ and $X(s)$ is given by

$$\frac{H(s)}{X(s)} = \frac{K}{Ts + 1 + K} \quad (4-7)$$

Since the Laplace transform of the unit-step function is $1/s$, substituting $X(s) = 1/s$ into Equation (4-7) gives

$$H(s) = \frac{K}{Ts + 1 + K} \frac{1}{s}$$

Expanding $H(s)$ into partial fractions gives

$$H(s) = \frac{K}{1 + K} \frac{1}{s} - \frac{K}{1 + K} \frac{1}{s + (1 + K)/T}$$

Taking the inverse Laplace transforms of both sides of this last equation, we obtain the following time solution $h(t)$:

$$h(t) = \frac{K}{1 + K} (1 - e^{-t/T_1}), \quad \text{for } t \geq 0 \quad (4-8)$$

where

$$T_1 = \frac{T}{1 + K}$$

The response curve $h(t)$ is plotted in Figure 4-7(d). From Equation (4-8), notice that the time constant T_1 of the closed-loop system is different from the time constant T of the feed-forward block.

From Equation (4-8), we see that as t approaches infinity the value of $h(t)$ approaches $K/(1 + K)$, or

$$h(\infty) = \frac{K}{1 + K}$$

Since $x(\infty) = 1$, there is a steady-state error of $1/(1 + K)$. Such an error is called *offset*. The value of the offset becomes smaller as the gain K becomes larger.

Offset is a characteristic of the proportional control of a plant whose transfer function does not possess an integrating element. (In such a case we need a nonzero error to provide a nonzero output.) To eliminate such offset, we must add integral control action. (Refer to Section 5-3.)

4-3 SECOND-ORDER SYSTEMS

In this section, we shall obtain the response of a typical second-order control system to a step input, ramp input, and impulse input. Here we consider a dc servomotor as an example of a second-order system. Conventional dc motors use mechanical brushes and commutators that require regular maintenance. Due to improvements that have been made in the brushes and commutators, however, many dc motors used in servo systems can be operated almost maintenance free. Some dc motors use electronic commutation. They are called brushless dc motors.